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Generic Galois Extensions and Problems in Field Theory

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INTRODUCTION

A course on elementary field theory usually includes the theory of Kummer extensions and Artin–Schreier polynomials. More advanced treatments include the theory of Witt vectors and abelian p extensions. Each of these theories contains, within it, a description of all Galois extensions with a fixed group, either cyclic or abelian. In this paper we will highlight some common properties of these descriptions and generalize them to more groups and fields. We also consider the consequences of such a description and thereby shed light on both Noether's Problem and the Grunwald–Wang theorem. Included in our results is an elementary proof of one of the known counterexamples to Noether's Problem. We also derive a generalization, to arbitrary valued fields and both abelian and nonabelian groups, of the Grunwald–Wang theorem. A weaker version of these results was announced in [11]. Contained in our results are the theorems of [9] and [18], where Grunwald–Wang theorems for abelian groups and discrete valued fields are proved. In particular, we have an elementary proof of a large chunk of the classical Grunwald–Wang theorem in number theory, which includes the (also elementary) results of [9] and [18], and which is strong enough to prove the cyclicity of division algebras over global fields.

It is helpful to begin with a well-known example. Let F be a field of characteristic p, and C_p the cyclic group of order p. Every Galois extension K/F with group C_p is gotten by adjoining to F a root of a polynomial of the form $y^p - y - a$, $a \in F$. Conversely, each polynomial $y^p - y - a$ yields a Galois extensions K/F, with group C_p, though K might not be a field.

Let us rewrite the above simple facts in a suggestive way. Set R to be the polynomial ring $F[x]$. Define S to be $R[y]/(y^p - y - x)$. Then S/R is a Galois extension of commutative rings with group C_p. Furthermore, every Galois extension K/F with this group is, in a well-defined way, a

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specialization of S/R. In fact, if K is any field containing F, and L/K is Galois with group C_p, then L/K is a specialization of S/R.

S/R is an example of what we will later define as a generic Galois extension. The generic extension S/R embodies the description of cyclic extensions contained in Artin–Schreier theory. The fact that R is a polynomial ring reflects the fact that the constant "a" in the polynomial $y^p - y - a$ can be chosen arbitrarily.

The description of abelian extensions in Kummer theory involves several variables. These variables are not, strictly speaking, independent but satisfy a condition of the form $s(a_1 \cdots a_n) \neq 0$ for a polynomial s. To illustrate, let F be a field of characteristic 0 containing a primitive nth root of one. Set G to be the abelian group $C_n \oplus C_n$, and let K be a field containing F. Any Galois extension L/K with group G is of the form $F(\alpha_1, \alpha_2)$, where α_i is a root of $y^n - a_i$. Conversely, the pair $y^n - a_1$, $y^n - a_2$, yield a Galois extension L/K (L not necessarily a field) exactly when $a_1 a_2 \neq 0$. In other words, if we set $R = F[x_1, x_2](1/x_1 x_2)$ and $S = R[y_1, y_2]/(y_1^n - x_1, y_2^n - x_2)$, then S/R is a generic Galois extension for G over F.

The two examples above serve to motivate the general definition of a generic Galois extension presented in Section 1. Also in Section 1 are the most elementary properties of generic Galois extensions. In Section 2, we consider abelian extensions and construct generic Galois extensions for abelian groups. These constructions are based on an observation by Albert [17, p. 221], and parallel the argument in [9] and [18]. In those cases where no such extensions exist, one cay apply the theory of tractable extensions as developed in the second half of Section 2. In Section 3, it is shown how in some cases generic extensions can be built from smaller generic extensions. Also in that section are some observations concerning the construction of regular extensions, which imply some simple facts about groups appearing over Hilbertian fields. In Section 4, generic extensions are constructed over fields of prime characteristic p. Finally, in Section 5, we attend to Noether's Problem, the Grunwald–Wang theorem and generalizations, and the lifting problem for Galois extensions over local rings. This last topic provides a sort of converse to all of this, namely, a property of Galois extensions which is equivalent to the existence of a generic one.

Many of the results of this paper were announced in [11]. The elementary proof of a counterexample to Noether's Problem is contained in Section 5 and one is independent from the rest. Combining Sections 1, 2, and 5 one can get the generalized Grunwald–Wang theorem announced in [11]. The main theorem of Section 4 was also stated in [11].

Let us begin with some notation. As above, C_n will denote the cyclic group of order n. In this paper, F will always be a field, and R and S commutative F algebras. If F has characteristic 0, we adopt the convention that this characteristic is prime to all integers $n \neq 0$. Given n prime to the

characteristic of F, we denote by $\rho(n)$ a fixed primitive nth root of one in some algebraic closure of F. Finally, $[R:F]$ denotes the dimension of R as a vector space over F.

Let $R[y]$ be the polynomial ring over R and $f(y) \in R[y]$ a monic polynomial. We denote by $R\{f(y)\}$ the F algebra $R[y]/(f(y))$. R imbeds naturally in $R\{f(y)\}$. The image of y in $R\{f(y)\}$ is called a canonical generator. If $\varphi: R \to S$ is an F algebra homomorphis, we denote by $M \otimes_\varphi S$ the tensor product of an R module M with S considered as an R module via φ. $R[M]$ will denote the symmetric algebra of M over R. Of course, if M is a free R module, $R[M]$ is just a polynomial ring. R^+ is the additive group of R, while R^* will denote the multiplicative group of units of R.

During most of this paper we will make use of the language and results of the Galois theory of commutative rings. We refer the reader to one of the excellent sources, for example [3], for any material missing here. What follows is something of a summary of that part of this Galois theory which we will use. Note that in all except 0.6, we do not use the fact R, S are F algebras.

Let G be a finite group acting faithfully on S, and assume R is a subalgebra of the fixed ring $S^G = \{s \in S \mid \sigma(s) = s \text{ for all } \sigma \in G\}$. There is a natural map, ψ, from the trivial crossed product $\Delta(S/R, G)$ into $\mathrm{Hom}_R(S, S)$. We say S/R is Galois with group G if ψ is an isomorphism (e.g. [3, p. 81]). Note that the Galois theory for S/R associates subgroups $H \subset G$ with subrings $T = S^H$. In this situation, S/T is Galois with group H. If H is normal, T/R is Galois with group G/H.

It is important to realize that the ring extension S/R does not determine the Galois group of S/R. Thus, in order to specify a G-Galois extension S/R, it is necessary to specify the action of G on S. Similarly, we will say two G-Galois extensions S/R and S'/R are isomorphic only if there is an isomorphism from S to S' which preserves the G actions.

If T/S and S/R are Galois, it is, of course, not true that T/R is Galois. But this is true if the Galois group of S/R extends to T.

LEMMA 0.1. *Let $T \supseteq S \supseteq R$ be such that T/S is Galois with group H and S/R is Galois with group N. For each $\sigma \in N$ assume that σ extends to an automorphism, called z_σ, of T. Suppose, further, that the following two properties hold.*

(a) *For any two $\sigma, \tau \in N$, $z_\sigma z_\tau$ is an automorphism of the form $z_\eta h$, for $\eta = \sigma\tau \in N$, and $h \in H$.*

(b) *For each $h \in H$ and each $\sigma \in N$, $z_\sigma h z_\sigma^{-1} \in H$.*

Then T/R is Galois with group G generated by H and the z_σ's.

Proof. By our assumptions, every element of G can be written as $z_\sigma h$ for

some $h \in H$ and $\sigma \in N$. The result now easily follows using [3, Proposition 1.2 (5), p. 81].

We next quote four results and give references instead of proofs.

PROPOSITION 0.2. *Let S/R be a Galois with group G.*

(a) [2, p. 13]. *Then S is projective as a module over the group ring $R[G]$. In particular, the group cohomology groups $H^n(G, S^+)$ are zero for $n \geqslant 1$.*

(b) [3, p. 116]. *Suppose R has zero Picard group. Then $H^1(G, S^*) = (0)$.*

(c) [13, p. 528]. *Let $R' \subset R$ be an F subalgebra. Then there are F algebras $R'' \subset S''$ such that $R' \subseteq R'' \subseteq R$ and*

(i) *S''/R'' is Galois with group G and $S \simeq S'' \otimes_{R''} R$.*

(ii) *R'' is finitely generated as an algebra over R'.*

(d) [3, p. 85]. *Let φ: $R \to R'$ be a ring homomorphism and set $S' = S \otimes_\varphi R'$. Then S'/R' is Galois with group G.*

Consider again the trivial crossed product $\Delta(S/R, G) = \Delta$. A Δ module, M, is easily seen to be just an S module with a semilinear G action. That is, for each $\sigma \in G$ there is a $\psi_\sigma: M \to M$ such that ψ_1 is the identity, $\psi_\sigma \circ \psi_\tau = \psi_{\sigma\tau}$, and $\psi_\sigma(sm) = \sigma(s)\,\psi_\sigma(m)$. We denote by M^G the fixed set $= \{m \in M \mid \psi_\sigma(m) = m$ for all $\sigma \in G\}$. Standard theory (e.g. [5, p. 44]) shows that $S \otimes_R M^G \cong M$ as Δ modules and, if M is S projective, M^G is R projective. Given such an M, we note that we have a G action on $S[M]$, and that $S[M]^G = R[M^G]$. Since $S[M] = R[M^G] \otimes_R S$, $S[M]/R[M^G]$ is Galois with group G.

We will have occasion to induce larger Galois extensions from smaller ones, as follows. Let S/R be Galois with group H and consider H as a subgroup of some finite group G. We view $R[H] \subset R[G]$. Write $\mathrm{Ind}_H^G(S) = R[G] \otimes_{R[H]} S$. As a module, $\mathrm{Ind}_H^G(S)$ is a direct sum of S's, each direct summand corresponding to a left coset of H in G. $\mathrm{Ind}_H^G(S)$ can therefore be made into an F algebra. In effect, we define $(\sigma \otimes s)(\tau \otimes t) = 0$ unless $\sigma^{-1}\tau \in H$, and $(\sigma \otimes s)(\sigma \otimes t) = (\sigma \otimes st)$. $\mathrm{Ind}_H^G(S)$ is a $R[G]$ module, and so has a G action. This action preserves the above multiplication. We claim:

PROPOSITION 0.3. *$\mathrm{Ind}_H^G(S)/R$ is Galois with group G.*

Proof. By [3, Proposition 1.2 (5), p. 31], it suffices to show that for any maximal ideal, M, of $\mathrm{Ind}_H^G(S)$ and each $1 \neq \sigma \in G$, there is an $x \in \mathrm{Ind}_H^G(S)$ with $\sigma(s) - s \notin M$. But the maximal ideals of $\mathrm{Ind}_H^G(S)$ are all of the form $S \oplus \cdots \oplus N \oplus \cdots \oplus S$, where $N \subset S$ is maximal. Without loss of generality,

we can assume M has the form $N \oplus S \oplus \cdots \oplus S$. If $\sigma \in H$, we use the fact that S/R is Galois. If $\sigma \notin H$, one can take $s = (1, 0,..., 0)$. Q.E.D.

Turning to the case $R = K$ is a field, let L/K be Galois with group G. We emphasize that here and in the rest of the paper, L is not necessarily a field. Of course, L is the direct sum of fields. In fact, it is easy to see that if $e \in L$ is a primitive idempotent, and $H = \{\sigma \in G \mid \sigma(e) = e\}$, then eL is a field Galois over K with group H and $L \cong \mathrm{Ind}_H^G(eL)$. H is called a decomposition group of L and G. Though not unique, every other decomposition group is a conjugate of H in G, and any such conjugate is a decomposition group.

When L is a field, the group G is determined by the extension L/K. It follows that if L_1/K, L_2/K are G Galois extensions with the L_i's fields, and $L_1 \cong L_2$ as K algebras, then L_1/K and L_2/K are isomorphic as G-Galois extensions. When the L_i's are not fields, a bit more has to be said.

LEMMA 0.4. *Let L_1/K and L_2/K be G Galois extensions such that $L_1 \cong L_2$ as K algebras. Suppose $H \subseteq G$ is a decomposition group for both L_1/K and L_2/K. Then $L_1 \cong L_2$ as G Galois extensions.*

Proof. Let $e_i \in L_i$ be primitive idempotents such that $H = \{\sigma \in G \mid \sigma(e_i) = e_i\}$. Set $L_i' = e_i L_i$. There is an isomorphism $\varphi' : L_1' \to L_2'$ which preserves the action of H. Clearly φ' extends to an isomorphism $\mathrm{Ind}_H^G(L_1') \to \mathrm{Ind}_H^G(L_2')$ which preserves the G actions. Since $L_i \cong \mathrm{Ind}_H^G(L_i')$ as G Galois extensions, we are done. Q.E.D.

Not assuming L is a field puts a slightly different flavor into the usual Galois theory over K. It is therefore necessary, in this next proposition, to record that in many instances the usual properties of Galois theory carry over.

PROPOSITION 0.5. *Let L/K and G be as above, and suppose G has order m.*

(a) $H^1(G, L^*) = (0)$; $H^n(G, L^+) = (0)$ *for all* $n \geqslant 1$.

(*b*) *The automorphisms in G are algebraically independent over K. That is, if f is any polynomial over K in m variables, then there is $a \in L$ such that $f(\sigma_1(a),..., \sigma_m(a)) \neq 0$, where $G = \{\sigma_1,..., \sigma_m\}$.*

(c) *L has a normal basis over K. That is, $L^+ \cong K[G]$ as $K[G]$ modules.*

(d) *Suppose K has characteristic prime to n, $\rho(n) \in K$, and $G = C_n$. Then $L \cong K\{y^n - a\}$ for some $0 \neq a \in K$. There is a canonical generator $\alpha \in L$ and a generator $\sigma \in C_n$ such that $\sigma(\alpha) = \alpha\rho(n)$.*

(e) *Suppose K has characteristic a prime, p, and $G = C_p$. Then $L \simeq K\{y^p - y - a\}$ for some $a \in K$. There is a canonical generator $\alpha \in L$ and a generator $\sigma \in C_p$ such that $\sigma(\alpha) = \alpha + 1$.*

Proof. Part (a) follows from 0.2(a) and (b). Parts (d) and (e) follow using (a), exactly as in the case L is a field. Parts (b) and (c) follows from exactly the same arguments that apply when L is a field (e.g. [4, p. 283]).

With (d) and (e) in mind, we return to Galois extensions of general commutative F algebras. We mention some easy results which will allow us to construct Galois extensions. We will omit any proofs as all follow by easy applications of [3, p. 81].

PROPOSITION 0.6. (a) *Let n be prime to the characteristic of F and assume $\rho = \rho(n) \in F$. If a is a unit of R, and $S = R\{y^n - a\}$, then S/R is Galois with group C_n. The action of C_n on S is defined by $\sigma(\alpha) = \alpha\rho$, where α is a canonical generator of S, and σ generates C_n.*

(b) *Let F have characteristic a prime, p. For any $a \in R$, set $S = R\{y^p - y - a\}$. Then S/R is Galois with group C_p. The action of C_p on S is defined by $\sigma(\alpha) = \alpha + 1$, where α is a canonical generator of S, and σ generates C_p.*

(c) *Let S_1/R, S_2/R be Galois with groups H_1 and H_2. Then $S_1 \otimes_R S_2/R$ is Galois with group $H_1 \oplus H_2$.*

1. THE DEFINITION

The first goal of this section is to formally define a generic Galois extension. Thereafter we will give some easy examples and present a few basic properties.

DEFINITION 1.1. Let G be a finite group and F a field. An extension of commutative F algebras, S/R, is a generic extension for G over F if and only if:

(a) R is of the form $F[x_1, ..., x_n](1/s)$.

(b) S/R is Galois with group G.

(c) Let K be a field containing F and let L/K be a Galois extension with group G. Then there is an F algebra map $\varphi: R \to K$ such that $L \cong S \otimes_\varphi K$, the isomorphism perserving the respective G actions.

The map $\varphi: R \to K$ we call a specialization. That $L \cong S \otimes_\varphi K$ we express by saying φ realizes L/K.

As stated in the Introduction, Kummer theory and Artin–Schrier theory provide examples of generic Galois extensions. This is the substance of the next theorem. Since direct sums will be handled in a general way, we restrict ourselves to cyclic groups in this theorem.

THEOREM 1.2. (a) *Let F be a field of characteristic a prime p. Set $R = F[x]$ and $S = R\{y^p - y - x\}$. Then S/R is generic for C_p over F.*

(*b*) *Let F have characteristic prime to n and assume $\rho(n) \in F$. Set $R = F[x](1/x)$ and $S = R\{y^n - x\}$. Then S/R is generic for C_n over F.*

Proof. In both (a) and (b), 1.1(a) of the definition is clear and 1.3(b) follows from 0.6. As for 1.1(c), first consider (a). If $K \supseteq F$ is a field and L/K is Galois with group C_p, then by 0.5(d) $L \cong K\{y^p - y - a\}$, for some $a \in K$. We define $\varphi: R \to K$ by setting $\varphi(x) = a$ and observe that φ realizes L/K. In part (b), let $K \supseteq F$ be a field and L/K Galois with group C_n. By 0.4(e), $L \cong K\{y^n - a\}$ for some $0 \neq a \in K$. Define $\varphi: R \to K$ by setting $\varphi(x) = a$, and note that φ realizes L/K. Q.E.D.

Next, we point out the easy fact that a generic extension for G over F yields a generic extension for G over any field $F' \supseteq F$.

LEMMA 1.3. *Let S/R be generic for G over F. Suppose F' is a field containing F. Then $F' \otimes_F S/F' \otimes_F R$ is generic for G over F'.*

Proof. Parts (a) and (b) of Definition 1.3 are trivial. As for (c), let K be a field containing F' and L/K a Galois extension with group G. Choose $\varphi: R \to K$ such that φ realizes L/K. φ induces a unique F' algebra map φ': $F' \otimes_F R \to K$ and φ' realizes L/K. Q.E.D.

Suppose K contains F, but K is now a finite direct sum of fields. Assume S/R is a generic Galois extension for G over F. We next observe that by specializing S/R one can also get all G Galois extensions of K.

PROPOSITION 1.4. *Let K, G, and S/R be as above. Assume L/K is Galois with group G. Then there is a specialization $\varphi: R \to K$ such that φ realizes L/K.*

Proof. Write $K = K_1 \oplus \cdots \oplus K_r$, each K_i being a field. Then $L = L_1 \oplus \cdots \oplus L_r$, where L_i/K_i is Galois with group G. Any F algebra morphism $\varphi: R \to K$ is determined by the induced maps $\varphi_i: R \to K_i$. Choose φ_i such that it realizes L_i/K_i. The induced map $\varphi: R \to K$ realizes L/K.

Q.E.D.

Finally, we close this section with the first of a series of results that yield generic Galois extensions from generic extensions for smaller groups. The following theorem concerns the direct sum of two groups. In Sections 3 and 4 more nontrivial results along these lines will be proved.

THEOREM 1.5. *Let F be a field. Assume H_1 and H_2 are finite groups with generic Galois extensions over F. Then $G = H_1 \oplus H_2$ has a generic Galois extension over F.*

Proof. Let S_i/R_i be a generic extension for H_i over F. Set $R = R_1 \otimes_F R_2$ and $S = S_1 \otimes_F S_2$. An easy verification shows that S/R is generic for G over F.

2. Abelian Extensions

Kummer Theory yields abelian generic Galois extensions over fields with enough roots of unity. The first goal of this section (Theorem 2.1) is to construct such extension over a much wider class of fields. In the second half of this section we consider fields and abelian groups left out of Theorem 2.1. We construct generic-like extensions for a subclass of all abelian Galois extensions. This subclass we call the tractable extensions. We present a series of results giving properties of tractable extensions, and showing that cyclotomic extensions are tractable. The purpose of this machinery is to be able to apply the techniques of Section 5, concerning lifting problems and Grunwald type theorems, to abelian extensions where no generic extension exists.

We begin by stating our main theorem on the existence of generic extensions for abelian groups. Let A be a finite abelian group, and F a field of characteristic prime to the order of A. Denote by r the highest power of 2 dividing the exponent of A.

THEOREM 2.1. *Suppose $F(\rho(r))/F$ is a cyclic extension. Then there is a generic Galois extension for A over F.*

The essence of Theorem 2.1 is a description of Galois extensions with cyclic Galois groups (Theorems 2.3 and 2.4). This description was first stated by Albert in the case of groups of odd prime order, was generalized in [9], and appears in full generality in [18]. Because of our different viewpoint, it seems wise to include a complete proof of this description here.

We begin the proof of 2.1 by observing that because of 1.5, we may assume that A is cyclic of prime power order $q = p^n$. That is, $A = C_q$. If q' is any odd prime power, then $F(\rho(q'))/F$ is always cyclic. Thus our assumption amounts to assuming $F(\rho(q))/F$ is cyclic. Set s to be the degree of $F(\rho(q))/F$. The Galois group of $F(\rho(q))/F$ is a subgroup of the group of units modulo q, so s divides $(p-1)\,p^{n-1}$. Let R be an F algebra. Set $\rho = \rho(q)$ and $R' = R \otimes_F F(\rho)$. By assumption, R'/R is a cyclic Galois extension. The Galois group of R'/R has a generator, τ, such that $\tau(\rho) = \rho^m$. The integer m has order s in the group of units modulo q. In what follows, it will be necessary to choose m with a special property.

LEMMA 2.2. *Suppose $m \not\equiv -1(q)$ or q is odd. Then one can choose m such that $k = (m^s - 1)/q$ is prime to q.*

Proof. k is prime to q if and only if m does not have order s modulo pq. If q is odd this follows from the fact that the group of units modulo pq, U_{pq}, is cyclic. If $p=2$, then the canonical map $U_{2q} \to U_q$ is of the form $C_2 \oplus C_q \to^f C_2 \oplus C_{q/2}$, where f is the direct sum of the identify map and the surjection $C_q \to C_{q/2}$. The integer -1, considered as an element of U_q, generates the C_2 factor. Lemma 2.2 now easily follows. Q.E.D.

Now let T be an F algebra, σ an automorphism of T, and u, v integers. For $a \in T$ a unit, define $M_\sigma^{u,v}(a) = a^{u^{v-1}}\sigma(a)^{u^{v-2}} \cdots \sigma^{v-1}(a)$. We mention two useful identities of this function. First of all, assume a, $z \in T$ satisfy $\sigma(a) = a^u z$. Then $\sigma^v(a) = \sigma^{v-1}(a^u z) = \sigma^{v-2}(a^{u^2} z^u \sigma(z)) = \cdots = a^{u^v} M_\sigma^{u,v}(z)$. Secondly, assume $\sigma^v(a) = a$. Then $\sigma(M^{u,v}(a)) = \sigma(a)^{u^{v-1}} \sigma^2(a)^{u^{v-2}} \cdots \sigma^v(a) = (M_\sigma^{u,v}(a))^u a^{-u^v+1}$. Our special interest is in the case $T = R'$, σ is τ, and q is odd or $m \not\equiv -1(q)$. We will always assume that m has been chosen as in 2.2. We call this Case I. Set $M_\tau = M_\tau^{m,s}$. We use the function M_τ to characterize C_q extensions.

THEOREM 2.3. *Assume the conditions of Case* I.

(a) *Suppose* $b \in (R')^*$, *and* $a = M_\tau(b)$. *Set* $S' = R'\{y^q - a\}$. *Then* $S' \cong S \otimes_R R'$, *where* S/R *is Galois with group* C_q.

(b) *Assume* $K = R$ *is field, set* $K' = R'$, *and suppose* L/K *is Galois with group* C_q. *Write* $L' = L \otimes_K K'$. *Then there is a* $b \in (K')^*$ *such that* $L' \cong K'\{y^q - a\}$, *where* $a = M_\tau(b)$.

Proof. (a) Let $\alpha \in S'$ be a canonical generator. Extend τ to S' by defining $\tau(\alpha) = \alpha^m b^{-k}$. We computed above that $\tau(M_\tau(b)) = M_\tau(b)^m b^{-kq}$. Then $\tau(\alpha)^q = \alpha^m b^{-kq} = M_\tau(b)^m b^{-kq} = \tau(a)$ and so τ is well defined. We also computed that $\tau^s(\alpha) = \alpha^{m^s} M_\tau(b)^{-k}$. But $\alpha^{m^s} M_\tau(b)^{-k} = \alpha a^k M_\tau(b)^{-k} = \alpha$. Thus τ has order s on S'. Denote by σ the automorphism of S' given by $\sigma(\alpha) = \alpha\rho$. We have $\sigma\tau(\alpha) = \sigma(\alpha^m b^{-k}) = \tau(\alpha\rho) = \tau\sigma(\alpha)$. Hence σ and τ commute. Using 0.1, we conclude that S'/R is Galois with a group generated by σ and τ. Set S to be the fixed ring of S' under τ. Then S/R is desired.

(b) Given L', L, K, and K' as in (b), τ extends to L' by defining it to be the identity on L. The Galois group, C_q, of L/K extends to L' in a similar manner, and commutes with τ. Since K' is a direct sum of fields containing $F(\rho)$, $L' \cong K'\{y^q - a\}$ for $a \in (K')^*$. Choose $\alpha \in L'$ a canonical generator and σ a generator of C_q such that $\sigma(\alpha) = \alpha\rho$. Then $\sigma(\tau(\alpha)) = \tau(\sigma(\alpha)) = \tau(\alpha\rho) = \tau(\alpha)\rho^m$. Hence $\tau(\alpha)/\alpha^m$ is fixed by σ and $\tau(\alpha) = \alpha^m z$ for z a unit in K'. Furthermore, $\alpha = \tau^s(\alpha) = \alpha^{m^s} M_\tau(z) = \alpha a^k M_\tau(z)$. Thus $a^k = M_\tau(z^{-1})$. Choosing k' such that $kk' \equiv 1(q)$ we have that $a = M_\tau(z^{-k'}) w^q$ for $w = a^{-r}$ and $r = (kk' - 1)/q$. Replacing α by αa^r, and setting $b = z^{-k'}$, we have $\tau(\alpha)/\alpha^m = zz^{rq} = b^{-k}$ and $a = M_\tau(b)$. Q.E.D.

At this point we must handle the case $m \equiv -1$ (q) and q is a power of 2. We have $\tau(\rho) = \rho^{-1}$, and we may assume $m = -1$. We call this Case II.

THEOREM 2.4. *In Case* II, *the following holds.*

(a) *Suppose* $b \in (R')^*$, $z \in R^*$, *and* $a = (\tau(b)/b)\, z^{q/2}$. *Set* $S' = R'\{y^q - a\}$. *Then* $S' \cong S \otimes_R F'$, *where* S/R *is Galois with group* C_q.

(b) *Assume* $K = R$ *is a field,* $K' = R'$, *and* L/K *is Galois with group* C_q. *Write* $L' = L \otimes_K K'$. *Then there is a* $b \in (K')^*$ *and* $z \in K'$ *such that* $L' \cong K'\{y^q - a\}$, *where* $a = (\tau(b)/b)\, z^{q/2}$.

Proof. (a) Since $\tau(a) = (b/\tau(b))\, z^{q/2}$ we have that $a\tau(a) = z^q$ or $\tau(a) = a^{-1}z^q$. If $\alpha \in S'$ is a canonical generator, we can define $\tau(\alpha) = \alpha^{-1}z$ and proceed exactly as in 2.3(a). Note that $\tau^2(\alpha) = \alpha$ because $\tau(z) = z$.

(b) Write $L' = K'\{y^q - a\}$ and choose $\alpha \in L'$ a canonical generator. As in 2.3(b), $\tau(\nu) = \alpha^{-1}z$, where $z \in K'$. But $\alpha = \tau^2(\alpha) = \alpha z^{-1}\tau(z)$ so $z \in K$. Taking qth powers, $\tau(\alpha) = a^{-1}z^q$. Since $z \in K$, we have that $(az^{-q/2})\, \tau(az^{-q/2}) = 1$. By 0.5, there is a unit $b \in K'$ such that $az^{-q/2} = \tau(b)/b$.

Q.E.D.

We are now in a position to define our generic Galois extensions. Consider, first or all, case I. Set $R' = F(\rho(q))[x_1, ..., x_s](1/t)$, where $t = x_1 \cdots x_s$. Extend τ to an automorphism of R' by setting $\tau(x_i) = x_{i+1}$, $\tau(x_s) = x_1$. Denote by R the fixed ring of R' under τ. In Case II, set $R' = F(\rho(q))[x_1, x_2, x_3](1/x_1x_2x_3)$. Define τ on R' by setting $\tau(x_1) = x_2$, $\tau(x_2) = x_1$, and $\tau(x_3) = x_3$.

LEMMA 2.5. (a) *Case* I: $R = F[z_1, ..., z_s](1/t)$ *for some* $z_1, ..., z_s \in R$.

(b) *Case* II: $R = F[z_1, z_2, x_3](1/x_1x_2x_3)$ *for some* $z_1, z_2 \in R$.

Proof. In Case I, denote by $V \subseteq R'$ the s dimensional space over $F(\rho(q))$ generated by $x_1, ..., x_s$. $\tau(V) \subset V$ and so τ is a semilinear action on V. V^τ, the fixed space, is an s dimensional vector space over F. Choosing $z_1, ..., z_s$ to be a basis for V^τ, the lemma is clear in this case. As for Case II, we set V to be the span of x_1 and x_2 and proceed in the same manner. Q.E.D.

We have $R' \cong F(\rho(q)) \otimes_F R$. In Case I, define $S' = R\{y^q - a\}$, where $a = M_\tau(x_1)$. In Case II, set $S' = R'\{y^q - a\}$, where $a = (\tau(x_1)/x_1)\, x_3^{q/2}$. By Theorems 2.3 and 2.4, τ extends to S' and we define S to be the fixed ring of τ on S'. A more precise version of 2.1 can now be stated.

PROPOSITION 2.6. *In either case,* S/R *is a generic Galois extension for* C_q *over* F.

Proof. 1.1(i) follows from Lemma 2.5, and 1.1(ii) is from Theorems 2.3

and 2.4. Suppose K is a field containing F and $K' = K \otimes_F F(\rho(q))$. Assume L/K is Galois with group C_q. For convenience, we only consider Case I as Case II is almost identical. If $L' = L \otimes_K K'$, then $L' \cong K'\{y^q - a\}$, where $a = M_\tau(b)$ for $b \in (K')^*$. Define $\varphi': R' \to K'$ by setting $\varphi'(x_1) = b$, $\varphi'(x_i) = \varphi'(\tau^{i-1}(x_1)) = \tau^{i-1}(b)$. φ' preserves the action of τ so φ' induces a map $\varphi: R \to K$. Using the description of the action of σ, τ given in the proof of 2.3, it is clear that $L' \cong S' \otimes_{\varphi'} K'$ as Galois extensions of K. Considering τ fixed subrings, we have that φ realizes L/K. Q.E.D.

In a later section the existence of a generic extension will imply a sort of Grunwald–Wang theorem, namely, that one can pull back extensions of a complete field to extensions of a dense subfield. But the classical Grunwald–Wang theorem yields some information about abelian extensions even when $F(\rho(q))/F$ is not cyclic. This suggests that something can be said about C_q extensions even when no generic extension exists. The rest of this section will develop such a theory: the theory of tranctable extensions. In essence, we construct a generic-like extension for all these tractable extensions, and then show that tractable extensions often exist.

So suppose F has characteristic prime to q and $F(\rho(q))/F$ is not cyclic for q a power of a prime. Then, of course, that prime must be 2 and q must be greater than or equal to 8. Let G be the Galois group of $F(\rho(q))/F$, so that G is a subgroup of the units group U_q. Since G is not cyclic, $G = C_2 \oplus C_s$, where $\sigma(\rho) = \rho^{-1}$ $(\rho = \rho(q))$ generates the C_2 direct summand of G. Choose τ to be a generator of C_s, and suppose $\tau(\rho) = \rho^m$. Choose m such that $k = (m^s - 1)/q$ is odd. Set F_1 to be the fixed field of σ in $F(\rho)$. Once again, let R be an F algebra and $R' = R \otimes_F F(\rho(q))$. Set $R_1 = R \otimes_F F_1$. Combining the arguments of Theorems 2.3 and 2.4 we conclude:

THEOREM 2.7. *Suppose $b \in (R')^*$, $z \in R_1^*$, and $a = M_\tau((\sigma(b)/b)\, z^{q/2})$. Set $S' = R'\{y^q - a\}$. Then the action of G on R' extends to S' and the fixed ring of G on S' is a Galois extension of R with group C_q.*

Proof. Let $\alpha \in S'$ be a canonical generator. Define $\tau(\alpha) = \alpha^m((\sigma(b)/b)\, z^{q/2})^{-k}$ and $\sigma(\alpha) = \alpha^{-1} M_\tau(z)$. One can easily check that σ, τ are well defined, $\sigma^2 = 1$, $\tau^s = 1$, and $\sigma\tau = \tau\sigma$. In this way we have defined an action of G on S'. Suppose η is in the Galois group of S'/R' such that $\eta(\alpha) = \alpha\rho$. Then $\eta\sigma(\alpha) = \eta(\alpha^{-1} M_\tau(z)) = \alpha^{-1}\rho^{-1} M_\tau(z) = \sigma(\alpha\rho) = \sigma\eta(\alpha)$. Similarly, $\eta\tau(\alpha) = \tau\eta(\alpha)$. Hence if S is the fixed ring of G on S', then η generates the Galois group of S/R. Q.E.D.

We use 2.7 to construct a generic looking extension for C_q over F. In fact, set $R' = F'[x_1, \ldots, x_{2s}, y_1, \ldots, y_s](1/t)$, where $t = x_1 x_2 \cdots x_{2s} y_1 \cdots y_s$. Extend τ to R' by defining $\tau(x_i) = x_{i+1}$ for $i \neq s$, $2s$, $\tau(x_s) = x_1$, $\tau(x_{2s}) = x_{s+1}$, $\tau(y_i) = y_{i-1}$ for $i < s$, and $\tau(y_s) = y_1$. Extend σ to R' by defining $\sigma(y_i) = y_i$,

$\sigma(x_i) = x_{i+s}$ for $i \leqslant s$, and $\sigma(x_i) = x_{i-s}$ for $i > s$. As before, the fixed ring of R' under τ and σ is of the form $F[z_1,..., z_{3s}](1/t)$. Call this fixed ring R. Define $a = M((\sigma(x_1)/x_1)\, y_1^{q/2})$ and set $S' = R'\{y^q - a\}$. By 2.7, $S' \cong S \otimes_R F'$, where S/R is Galois with group C_q. S/R is not, in general, generic for C_q over F. We will later show, for example, that if F is the rational field, then no generic extension for C_q can exist over F. However, if $K \supseteq F$ is a field, then some C_q extensions of K may be specializations of S/R. We say such extensions are tractable over F, and we call S/R the generic tractable extension for C_q over F.

The next series of results will display some basic properties of tractable extensions. Ostensibly, the definition of tractable depends upon F and our choice of m and τ. That this dependence is not serious is one of the consequences of the next theorem. Part (d) of the next result shows that tractable extensions generalize the extensions described in 2.1. In a theorem to come, we will show that there are tractable extensions over all fields such that $K(\rho)/K$ is not cyclic. Set $K_1 = F_1 \otimes_F K$, $K' = F' \otimes_F K$.

THEOREM 2.8. *Assume $F(\rho(q))/F$ is not cyclic and σ, τ are as above.*

(a) *Let L/K be a Galois extension with group C_q. L/K is tractable if and only if $L \otimes_K K' \cong K'\{y^q - a\}$ and there is a canonical generator, α, with the property that $z = \tau(\alpha)/\alpha^m$ has σ norm a qth power in K_1.*

(b) *With another choice of m or τ, the concept of tractability is unchanged.*

(c) *Let F_0 be a field such that $F \subset F_0 \subset K$ and $F_0(\rho)/F_0$ is not cyclic. L/K is tractable with respect to F if and only if it is tractable with respect to F_0.*

(d) *If $K(\rho)/K$ is cyclic then all C_q extensions of K are tractable.*

Proof. (a) Suppose z is as given. Arguing as in the proof of 2.3 we may assume $z = b^{-k}$, where $\alpha^q = a = M_\tau(b)$. Of course, $b\sigma(b)$ is also a qth power, say, d^q. Now $\tau(\alpha\sigma(\alpha))/(\alpha\sigma(\alpha))^m = z\sigma(z) = d^{-kq}$. That is, if $\beta = \alpha\sigma(\alpha)$, then $\tau(\beta) = \beta^m d^{-kq}$. Thus $\beta = \tau^s(\beta) = \beta^{ms} M_\tau(d^{-kq})$ or $\beta^{kq} = M_\tau(d^{kq})$. We have $\beta^k = M_\tau(d^k)\,\delta$, for δ a qth root of one. But as β, d are fixed by σ, δ is fixed by σ and so $\delta = \pm 1$. Changing d to $-d$ if necessary ($b\sigma(b) = d^q$ is still true), we have $\beta^k = M_\tau(d^k)$. If $kk' = 1 + rq$ then $\alpha\sigma(\alpha) = \beta = M_\tau(d^{kk'})\,\beta^{-rq} = M_\tau(d^{kk'})(a\sigma(a))^{-r} = M_\tau(d^{kk'})(M_\tau(d^{-rq})) = M_\tau(d)$. Finally, since $b\sigma(b) = d^q$, $b = (\sigma(c)/c\, d^{q/2})$ for some $c \in (K')^*$. All in all, we have $a = M_\tau((\sigma(c)/c)\, d^{q/2})$, $\sigma(\alpha) = \alpha^{-1} M_\tau(d)$, and $\tau(\alpha) = \alpha^m z = \alpha^m b^{-k} = \alpha^m((\sigma(c)/c)\, d^{q/2})^{-k}$. It is now clear that L/K is a specialization of S/R.

(b) Suppose, first of all, that τ is unchanged but that we have made another choice of m, say, m', where m' satisfies 2.2. Since $m \equiv m'(q)$, $(\tau(\alpha)/\alpha^m)(\tau(\alpha)/\alpha^{m'})^{-1}$ is a qth power. Applying part (a) yields this case.

Another choice, τ', of τ must be of the form τ^n or $\tau^n\sigma$, where n is prime to 2. Set $m' = m^n$ or $-m^n$, respectively. In the case $\tau' = \tau^n$, $\tau'(\alpha')/(\alpha')^{m'}$ is a product of $z = \tau(\alpha)/m$ and its conjugates. Once again, this case is clear by part (a).

Given what we have already done, it suffices to show tractability with respect to τ implies tractability with respect to $\sigma\tau$. Let L/K, K' and $L' \cong K'\{y^q - a\}$ be as in part (a) and assume L/K is tractable with respect to τ. We know that L' has a canonical generator, α, such that $z = \tau(\alpha)/\alpha^m$ has the form $((\sigma(b)/b)\, d^{q/2})^{-k}$ and that $\alpha\sigma(\alpha) = M_\tau(d)$, where σ fuxes d. Set

$$\alpha' = \alpha d^{-m^{s-1}}\tau^2(d)^{-m^{s-3}}\tau^4(d)^{-m^{s-5}} \cdots \tau^{s-2}(d)^{-m}.$$

Now α' is also a canonical generator for L'/K' and one can calculate that $\sigma\tau(\alpha')/(\alpha')^{-m}$ is equal to $\sigma(\tau(\alpha)/\alpha^m)$. Using (a) a third time, we are done.

(c) Since $F_0(\rho)/F_0$ is not cyclic, it's Galois group is generated by τ^e and σ, where e is a power of 2 less than s. Set $\tau' = \tau^e$, $m' = m^e$, and $s' = s/e$. Then τ' has order s', $\tau'(\rho) = \rho^{m'}$, and $k = (m'^{s'} - 1)/q$ is still odd.

We begin by exploring the relationship between $L' = L \otimes_{F_0} F_0(\rho)$ and $L'' = L \otimes_F F(\rho)$. Firstly, L'' is isomorphic to a direct sum $L' \oplus \cdots \oplus L'$ (e times). Identifying L'' with this direct sum in the proper way, we can described the actions of C_q, σ, and τ on L'' as follows. If $(a_1,..., a_e) \in L''$, then $\sigma(a_1,..., a_e) = (\sigma(a_1),..., \sigma(a_e))$ and $\tau(a_1,..., a_e) = (a_2,..., a_e, \tau'(a_1))$. If $\eta \in C_q$ then $\eta(a_1,..., a_e) = (\eta(a_1),..., \eta(a_e))$. Thus if $K' = K \otimes_{F_0} F_0(\rho) \subset L'$ and $K'' = K \otimes_F F(\rho) \subset L''$, we have identified K'' with $K' \oplus \cdots \oplus K'$. Denote by ρ'' the image of ρ in K'', and by ρ' the image of ρ in K'. Since $\tau(\rho'') = (\rho'')^m$, ρ'' has the form $(\rho', (\rho')^m,..., (\rho')^{m^{s'-1}})$.

Suppose L/K is tractable with respect to F. Choose $\alpha = (\alpha_1,..., \alpha_e) \in L''$ a canonical generator such that $\tau(\alpha)/\alpha^m = z$ has σ norm a qth power. Note that α_1 is a canonical generator for L'/K'. $\tau'(\alpha)/\alpha^{m'}$ is a product of z and its conjugates, and so also has σ norm a qth power. But the first component of $\tau'(\alpha)/\alpha^{m'}$ is just $\tau'(\alpha_1)/\alpha_1^{m'} = z_1$ and so z_1 has σ norm a qth power. We conclude that L/K is tractable with respect to F_0.

Conversely, suppose L/K is tractable with respect to F_0. Choose a canonical generator $\alpha_1 \in L'$ such that $\tau'(\alpha_1)/\alpha_1^{m'}$ has σ norm a qth power, say, d^q. Set $\alpha_j = (\alpha_1)^{m^{j-1}}$, and $\alpha = (\alpha_1,..., \alpha_e)$. If $\eta \in C_q$ satisfies $\eta(\alpha_1) = \rho'\alpha_1$, then $\eta(\alpha) = (\rho', \rho'^m,..., \rho'^{m^{e-1}})(\alpha_1,..., \alpha_e) = \rho''\alpha$. It follows that α is a canonical generator for L''/K''. But $\tau(\alpha)/\alpha^m$ has σ norm equal to $(1,..., 1, d^q)$, a qth power, and so L/K is tractable with respect to F.

(d) Finally getting to (d), we define F_2 to be the fixed field of τ on $F' = F(\rho)$. The Galois group of $K(\rho(q))/K$ is a subgroup of G, and is cyclic, and so must be generated by an element of the form τ^n, $\tau^n\sigma$, or σ.

In the first case, K must contain F_2. Using 2.6, define S_2/R_2 to be a generic extension for C_q over F_2 using 2.6. It suffices to construct an F

algebra map $\varphi_2: R \to R_2$ such that $S_2 \cong S \otimes_\varphi R_2$. Changing notation, write $R_2' = F'[v_1,..., v_s](1/v_1 \cdots v_s)$, where $R_2' = R_2 \otimes_{F_2} F'$. Let τ act on R_2' as in 2.6. If $R' = F'[x_1,..., x_{2s}, y_1,..., y_s](1/t)$ is as above, define $\varphi_2': R' \to R_2'$ by setting $\varphi_2'(x_i) = v_{i-s}$ for $s < i \leqslant 2s$, $\varphi_2'(x_i) = 1$ for $i \leqslant s$, and $\varphi_2'(y_i) = 1$ for all i. φ_2' preserves the action of τ and so restricts to a map $\varphi_2: R \to R_2$. $\varphi_2'(M_\tau((\sigma(x_1)/x_1)\, y_1^{q/2})) = M_\tau(v_1)$, so φ_2 induces an isomorphism $S_2 \cong S \otimes_{\varphi_2} R_2$.

Next, we consider the case that σ generates the Galois group of $K(\rho(q))/K$. Then $K \supseteq F_1$ and we can define a generic extension S_1/R_1 for C_q over F_1 using 2.6. Write $R_1' = R_1 \otimes_{F_1} F' = F'[v_1, v_2, v_3](1/v_1 v_2 v_3)$. Define φ_1': $R' \to R_1'$ by setting $\varphi_1'(x_s) = v_1$, $\varphi_1'(x_{2s}) = v_2$, $\varphi_1'(y_s) = v_3$, and $\varphi_1'(x_i) = 1 = \varphi_i(y_j)$ in all other cases. Note that φ_1' preserves σ and so induces a map φ_1: $R \to R_1$. Once again, a calculation shows that $S \otimes_{\varphi_1} R_1 \cong S_1$ as Galois extensions.

The last case is that $\tau^n\sigma$ generates $K(\rho(q))/K$. Let F'' be the fixed field of τ^n and σ on F'. By (c) above, L/K is tractable over F''. Now working over F'', we can use part (b) above to replace $\tau^n\sigma$ by τ^n, and apply the first case. This theorem is proved. Q.E.D.

Note that because of 2.8, one can speak of tractable extensions L/K without reference to F. The next result allows one to construct tractable extensions in the situation not covered by 2.8(d). Specifically, we show that cyclotomic extensions are tractable.

THEOREM 2.9. *Let $K(\rho(q))/K$ be a noncyclic extension where q is a power of 2. Define $\sigma: K(\rho(q)) \cong K(\rho(q))$ as above. If L is the fixed ring of σ on $K(\rho(q))$, then L/K is tractable.*

Proof. Let s be the order of L/K, and set $\delta = \rho(q)^{q/s}$. Of course, δ is a primitive sth root of one. We may assume $K(\delta)/K$ is not cyclic. Consider the automorphism $\tau: K(\rho) \cong K(\rho)$, where $\tau(\rho) = \rho^m$ and m is as in 2.2. Then τ generates the Galois group of L/K and has order s on $K(\rho)$. We can choose τ such that the integer m is a $q/4s$ power modulo q ($q/4$ is the exponent of the full group of units modulo q). Hence $m \equiv 1$ (q/s) or $ms \equiv s(q)$. Set $t = q/s$ and $\mu = \rho^s$. Then $\tau(\mu) = \rho^{ms} = \rho^s = \mu$ so $K(\mu)/K$ has degree 2. If $t \geqslant s$ then $K(\delta) \subset K(\mu)$ and so $K(\mu)/K$ is cyclic, a contradiction. Thus $s > t$. Modulo t, m must have order $s/t = n$. That is, τ has order n on $K(\delta)$. In all, $K(\delta)/K$ has order $2n = 2s/t$ and $K(\rho)/K(\delta)$ has order t. τ^n generates the Galois group of $K(\rho)/K(\delta)$.

Consider $L' = L \otimes_K K(\delta)$. L' has three automorphisms we now define. The automorphism $1 \otimes \tau$ we call σ', $\tau \otimes 1$ we call η, and $1 \otimes \tau$ we rename as τ'. If we set $K'' = K(\delta)$, then L' has the form $K'' \oplus K'' \oplus \cdots \oplus K''$ (n times). Considering L' as this direct sum, we describe the actions of η, τ, and σ. $\eta(a_1,..., a_n) = (\tau(a_n), \tau(a_1),..., \tau(a_{n-1}))$; $\tau(a_1,..., a_n) = (a_2,..., a_n, a_1)$; and

$\sigma'(a_1,..., a_n) = (\sigma(a_1),..., \sigma(a_n))$. $K(\delta) \subseteq L'$ has been identified with $\{(a, \tau(a),..., \tau^{n-1}(a)) \ a \in K(\delta)\}$, and $L \subset L'$ has been identified with $\{(a,..., a) \mid a \in L\}$.

Since m was chosen a $q/4s$ power modulo q, we have $m^n \equiv 1(s)$. As $(m^s - 1)/q$ is odd, one concludes that $c = (m^n - 1)/s$ is odd. Choose c' such that $cc' \equiv 1(q)$ and set $\rho' = \rho^{c'}$. Then $\eta^n(\rho') = \rho'\mu$. Now setting $\alpha = (\rho', \eta(\rho')\eta(\delta)^{-1},..., \eta^{n-1}(\rho')\eta^{n-1}(\delta)^{-n+1})$, we compute that $\eta(\alpha) = \alpha(\delta, \tau(\delta),..., \tau^{n-1}(\delta))$. In other words, α is a canonical generator for L' over $K(\delta)$. Consider $z = \tau'(\alpha)/\alpha^m$. z is an n-tuple all of whose entries are powers of δ. Hence the σ' norm of z is 1. We are done by 2.8(a). Q.E.D.

Let δ be, as above, a primitive s root of 1, where s is a power of 2 and $K(\delta)/K$ is not cyclic. We claim that one can always find a q such that the σ fixed field of $K(\rho(q))/K$ has order s. That is, we claim K always has at least one tractable C_s extension which is a field.

PROPOSITION 2.10. *Let s be a power of 2 and $\delta = \rho(s)$. Assume $K(\delta)/K$ is not cyclic. Then there is a power of 2, q, such that $K(\rho(q))/K$ has order $2s$.*

Proof. Let $K_{\#}$ be the field gotten by adjoining, to K, all 2^n roots of 1, $n \geqslant 2$. The Galois group of $K_{\#}/K$ is a subgroup of the group of units of the 2-adic integers. This group of units has exactly one nontrivial torsion element. Since $K_{\#}/K$ has degree greater than 2, it must be an infinite extension. Thus one can find a $q = 2^n$ such that $K(\rho(q))/K$ has degree greater than or equal to $2s$. Taking a power of $\rho(q)$ yields the proposition. Q.E.D.

The cyclotomic extensions studied in 2.9 have the property that there is a $z = \tau(\alpha)/\alpha^m$ such that z has σ norm 1. We call these strongly tractable extensions. This final corollary is an easy exercise.

COROLLARY 2.11. *Suppose K is a field and q a power of 2 such that K has characteristic prime to 2 and $K(\rho(q))/K$ is not cyclic. Assume L/K is a strongly tractable C_q extension. Let $q' > q$ be another power of 2 and view $C_q \subset C_{q'}$. Then $\mathrm{Ind}_{C_q}^{C_{q'}}(L/K)$ is strongly tractable.*

3. SOME CONSTRUCTIONS

Our goal, in this section and the next, is to construct generic Galois extensions for as many groups as we can. In this context, the best imaginable theorem might read as follows. If G, G' have generic extensions over F, then every extension of G by G' has a generic extension. In fact, in almost all circumstances (the exception is in Section 4) we cannot begin to attack this problem unless we restrict ourselves to split extensions. A few results along these lines are presented in this section. The techniques we will develop will

also apply to a different, but related, problem. Namely, we make some observations about constructing Galois extensions over Hilbertian fields.

Let us begin with a result which reverses the usual order of things. Given a generic extension for G, we ask when an image of G has a generic extension.

THEOREM 3.1. *Let F be a field and G a group with a generic Galois extension over F. Let $N \subset G$ be a normal subgroup.*

(a) *Suppose, for every field $K \supseteq F$ and every G/N extension L/K, L extends to a G Galois extension L'/K. Then there is a generic extension for G/N over F. (Note: L' need not be a field.)*

(b) *Suppose the epimorphism $G \to G/N$ splits. Then there is a generic extension for G/N over F.*

Proof. (a) Let S'/R be the generic Galois extension for G over R, and let S be the fixed ring of N in S'. Given any L/K as in (a), choose an extension L'/K such that L'/K has Galois group G and L is the fixed ring of N. If $\varphi: R \to K$ realizes L'/K' then considering S/R, φ realizes L/K. Hence S/R is a generic Galois extension for G/N over F.

(b) As $G \to G/N$ splits, there is a subgroup $H \subset G$ such that $H \cap N = (1)$ and $HN = G$. Of course, $H \cong G/N$. Suppose L/K is Galois with group G/N. Let us identify the Galois group of L/K with H. Set $L' = \mathrm{Ind}_H^G(L)$ (see 0.3). L'/K us Galois with group G. The fixed ring of N in L' is seen to be isomorphic to L. Thus L/K extends to a G Galois extension. We are done using part (a). Q.E.D.

For the rest of this section, we will present results which reverse the order of Theorem 3.1. That is, we will construct generic extensions for groups G out of generic extensions for normal subgroups N and quotients G/N. As mentioned before, we can most often do this when the map $G \to G/N$ splits.

In the constructions of this section, it will often be necessary to assume that for some generic Galois extension S/R, S is free as an R module. It is shown in Corollary 5.4 that if there is a generic Galois extension for G over F, then there is such an extension S/R such that S is free as an R module. Examining the proof of 5.4, we see that one can use 5.4 in this section without circularity.

The freeness of S over F is of interest to us because of the following result. Let S/R be Galois with group G and set $\Delta = \Delta(S/R, G)$ to be the trivial crossed product.

LEMMA 3.2. (a) *Let P be the S free module with basis $\{x_g \mid g \in G\}$. Define, for $h \in G$ and $\sum s_g x_g \in P$, $h(\sum s_g x_g) = \sum h(s_g)\, x_{hg}$. Then this makes P a Δ module isomorphic to Δ.*

(b) *Suppose $P = \Delta \oplus \cdots \oplus \Delta$ is a finitely generated free Δ module. Assume S is free as an R module. Form the symmetric algebra $S[P]$ and let G act on $S[P]$ by acting on S and P. Then the G fixed subring is a polynomial ring $R[x_1, ..., x_m]$.*

Proof. Part (a) is just the definition of Δ. As for (b), note that the G fixed elements of $S[P]$ form the symmetric algebra $R[P^G]$. This it suffices to assume $P = \Delta$ and it suffices to show Δ^G is a free R module. But $\Delta^G = \{\sum_{h \in G} h(x)\, x_h \mid s \in S\}$ and so, as R modules, $\Delta^G \cong S$. Q.E.D.

Before proceedings further, let us recall the basic facts about semidirect products of groups. Let G be a finite group and N a normal subgroup. As mentioned before $G \to G/N$ splits if and only if G has a subgroup H such that $H \cap N = (1)$ and $HN = G$. H is called a complement of N in G. There is a homomorphism $\psi\colon H \to \mathrm{Aut}(N)$ defined by $\psi(h)(n) = hnh^{-1}$. Equivalently, there is an action of H on N; $h \cdot n = hnh^{-1}$. N, H, and ψ determine G. In fact, G is isomorphic to the group consisting of ordered pairs $\{(n, h) \mid n \in N, h \in H\}$ subject to the operation $(n_1, h_1) * (n_2, h_2) = (n_1(\psi(h_1)(n_2)), h_1 h_2)$. We write this later group as $N \rtimes_\psi H$ or $N \rtimes H$ (ψ understood). It is called the semidirect product of N and H with respect to ψ.

A subclass of semidirect products, called wreath products, will be of special importance. Let M, G be finite groups. Set $N = \bigoplus_{\sigma \in G} M_\sigma$, where each M_σ is a copy of M. G acts on N by permuting the direct summands in the usual way. $N \rtimes G$ is called the wreath product of M with G, and is written $M \wr G$. Consider $G \subseteq M \wr G$ as the canonical complement of N.

THEOREM 3.3. *Suppose that there are generic Galois extensions for M and G over F. Then there is a generic Galois extension for $M \wr G$ over F.*

Proof. Suppose S'/R' is generic for G, and S''/R'' is generic for M, over F. Using 5.4, we assume S' is free as a module over R'. Consider $S_1 = \bigotimes_{\sigma \in G} S''_\sigma$, where the tensor product is over F and each S''_σ is a copy of S''. If we set $R_1 = \bigotimes_{\sigma \in G} R''_\sigma$, then S_1/R_1 is Galois with group $N = \bigoplus_{\sigma \in G} M_\sigma$. In addition, we can define an action of G on S_1 by permutation of the tensor product. Taken together, we have a well-defined action of $M \wr G$ on S_1. Set $S = S_1 \otimes_F S'$. Define an action of $M \wr G$ on S' by letting G act as given and letting N act trivially. In all, an action of $M \wr G$ on S has been defined. Using 0.1, S/R is Galois with group $M \wr G$. The claim is that S/R is a generic Galois extension for $M \wr G$ over F. We prove this claim by first describing R more precisely.

The fixed ring of N on S is clearly $R_1 \otimes_F S'$. R is the fixed ring of G on $R_1 \otimes_F S'$. Write each R''_σ as $F[x_{1,\sigma}, ..., x_{n,\sigma}](1/t_\sigma)$. Then one can describe $R_1 \otimes_F S'$ as the polynomial ring $S'[x_{i,\sigma} \mid 1 \leqslant i \leqslant n,\ \sigma \in G](1/s)$, where s is the product of al the t_σ's. With $R_1 \otimes_F S'$ written in this form, the action of G

can be described as follows. G acts as usual on S' and $\sigma'(x_{i,\sigma}) = x_{i,\sigma'\sigma}$. We can describe $R_1 \otimes_F S'$ in yet another way. Set P to be the free S' module of rank n, and $Q = \Delta(S'/R', G) \otimes_{S'} P$. By 3.2(a), $R_1 \otimes_F S'$ is the symmetric algebra $S'[Q](1/s)$, with G action induced by the G action on S' and Q. By 3.2(b), R has the form $R'[y_1, ..., y_m](1/s)$.

In order to show that every $M \wr G$ Galois extension is a specialization of S/R, we make the following observation, whose proof is an easy exercise in Galois theory.

LEMMA 3.4. *Let K be a field and L/K a Galois extension with group $M \wr G$. Denote by K' the fixed ring of N in L. There is a subalgebra $L' \subset L$ such that L'/K' is Galois with group M, and such that the multiplication map defines an isomorphism $L \cong \bigotimes_{\sigma \in G} \sigma(L')$, the product being over K'.*

Now if L/K is as in the above lemma, we choose φ': $R' \to K$ such that φ' realizes K'/K, and φ'': $R'' \to K'$ such that φ'' realizes L'/K'. Define φ: $R \to K$ as follows. Considering the R''_σ as subrings of R_1, and identifying R'' with R''_e ($e \in G$ the identity), we have $\sigma(R'') = R''_\sigma$. Denote by $(\varphi'')_\sigma$: $R''_\sigma \to K'$ the map $\sigma \circ \varphi \circ \sigma^{-1}$. Tensoring the $(\varphi'')_\sigma$'s defines a map φ_1: $R_1 \to K'$. φ_1 preserves the action of G and so $\varphi_2 = \varphi' \otimes \varphi_1$: $R' \otimes R_1 \to K'$ preserves G. Restricting to G fixed subrings we have φ: $R \to K$. By tracing back one sees that φ realizes L. This finishes the proof of Theorem 3.3. Q.E.D.

By combining 3.1 and 3.3, one can construct generic extensions for a wider class of semidirect products. Suppose $G = M \rtimes H$, and M is abelian. G is then an image of the wreath product $M \wr H$. In fact, if $N = \bigoplus M_\sigma$, then define f: $N \to M$ as follows. Each element of N is of the form $(a_\sigma)_{\sigma \in H}$, $a_\sigma \in M$. Set $f((a_\sigma)_{\sigma \in H}) = \prod_{\sigma \in H} \sigma^{-1}(a_\sigma)$. Then f is a surjection and preserves the action of H on N and M. Letting f be the identity on H, f extends to a surjection f: $M \wr H = N \rtimes H \to M \rtimes H$. Now define f': $M \to N$ by $f'(a) = (\sigma(a))_{\sigma \in H}$. f' also preserves H and so extends to a map $M \rtimes H \to M \wr H$. On M, $f(f'(a)) = a^r$, where r is the order of H. Thus if r is prime to the order of M, the map f splits. Combining 3.1 and 3.3 we have proved the following.

THEOREM 3.5. *Let M and H be finite groups of relatively prime order, and assume M is abelian. Suppose that there are generic Galois extensions for M and H over F. Then there is a generic Galois extension, over F, for any semidirect product $M \rtimes H$.*

In the special case M is cyclic and F has enough roots of unity, Theorem 3.5 can be improved. Our method of proof will be very similar to that of 2.3 and 2.4. In particular, we will use the maps M_τ defined in Section 2. It should be noted, however, that our argument also fits somewhat into the framework of 3.5. Our semidirect products are, of course, images of wreath products. However, the surjection involved does not, in general, split.

One way of viewing the argument to come is that we are proving the extension property of 3.1(a).

We begin with the case that M has prime power order. Let H be any finite group, C_q a cyclic group of prime power order q, and $\psi: H \to \mathrm{Aut}(C_q)$ a homomorphism with cyclic image. Form the semidirect product $G = C_q \rtimes H$. Choose $\tau \in H$ such that $\psi(\tau)$ generates $\psi(H)$. If σ is a generator for C_q, then $\psi(\tau)(\sigma) = \sigma^{m'}$ for m' relatively prime to q. Equivalently, considering $H \subset G$, we have $\tau^{-1}\sigma\tau = \sigma^m$, where $mm' = 1(q)$. Set $M_\tau = M_\tau^{m,s}$ and recall that if τ acts on a ring T, and $x \in T$, then $M_\tau^{m,s}(x) = x^{m^{s-1}}\tau(x)^{m^{s-2}} \cdots \tau^{s-1}(x)$.

We will use the M_τ function to construct a generic Galois extension for G. As in 2.3 and 2.4, there are two cases to be considered. Let s be the order of m modulo q. That is, s is the order of $\psi(H)$. If q is odd or $m \not\equiv -1(q)$, we can choose m such that $k = (m^s - 1)/q$ is prime to q (Case I). We call it Case II when $m = -1$ and q is a power of 2. We can describe Galois extensions with group G as follows. We do not include a proof of the following theorem, because the proof is almost exactly the same as the proofs of 2.3 and 2.4.

THEOREM 3.6. *Let R and K be F algebras, with K a field. Assume $\rho(q) \in F$.*

(a) *Let T/R be a Galois extension with group H. In Case* I, *suppose $b \in T$ is a unit fixed by $H' = \mathrm{kernel}(\psi)$ and set $a = M_\tau^{m,s}(b)$. Set $S = T\{y^q - a\}$, and let $\alpha \in S$ be a canonical generator. Extend the action of H to S by setting $\tau(\alpha) = \alpha^m b^{-k}$ and letting H' act trivially. Then we have that S/R is Galois with group G. In Case* II, *suppose $b \in T$, $c \in R$, are units. Set $a = (\tau(b)/b)\, c^{q/2}$, $S = T\{y^q - a\}$, and let $\alpha \in S$ be a canonical generator. The action of H extends to S by setting $\tau(\alpha) = \alpha^{-1}c$ and $\eta(\alpha) = \alpha$ if $\eta \in H'$. Using this action, S/R is Galois with group G.*

(b) *Suppose L/K is Galois with group G and let L' be the fixed ring of C_q. In Case* I, *there is a unit $b \in L'$, such that b is fixed by H', and $L \cong L'\{y^q - a\}$, where $a = M_\tau^{m,s}(b)$, and G acts as above. In Case* II, *there are units $b \in L'$, $c \in K$ such that $L \cong L'\{y^q - a\}$, where $a = (\tau(b)/b)\, c^{q/2}$, and G acts as above.*

Just as in Section 2, the above theorem is a sort of prescription for constructing a generic Galois extension for G. First of all, assume Case I. Let T'/R' be a generic Galois extension for H over F. Using 5.4, we may assume T' is free as an R' module. Form the localized polynomial ring $T = T'[x_1, \ldots, x_s](1/t)$, where $t = x_1 \cdots x_s$. We let H act on T by letting it act as given on T' and by defining $\tau(x_i) = x_{i+1}$ (indices mod s) and $\eta(x_i) = x_i$ if $\eta \in H'$. Set R to be the fixed ring of H in T. Lemma 3.2 implies that R has the form $R'[y_1, \ldots, y_s](1/t)$. Set $S = T\{y^q - a\}$, where $a = M_\tau(x_1)$. Considering 3.6, it is clear that S/R is a generic Galois extension for G over F. Using a parallel argument in Case II, we have the following theorem.

THEOREM 3.7. *Let C_q be the cyclic group of prime power order q, H a finite group, and $\psi: H \to \mathrm{Aut}(C_q)$ a homomorphism with cyclic image. Assume $\rho(q) \in F$ and that there is a generic Galois extension for H over F. Then there is a generic Galois extension for $C_q \rtimes H$ over F.*

Before expanding on 3.7, we consider in a more general context the machinery we will use. The constructions of generic extensions in this section all have the following form.

DEFINITION 3.8. Let N, H, G be finite groups with $N \subseteq G$ and $G/N = H$. Let T'/R' be a generic Galois extension for H over F, and suppose that R, S are R' algebras. We call S/R a generic ring extension of T'/R' for G, if the following properties hold.

(a) R is of the form $R'[x_1, ..., x_n](1/s)$.

(b) If $T = T' \otimes_{R'} R$, then $S \supseteq T \supseteq R$ is such that S/R is a Galois extension with group G and T is the fixed ring of N.

(c) Suppose K is a field containing F and L/K is Galois with group G. Call L' the fixed ring of N in L. Then any specialization $\varphi': R' \to K$ which realizes L'/K can be extended to a specialization $\varphi: R \to K$ which realizes L/K.

Of course, if S/R is a generic ring extension, as above, then S/R is a generic Galois extension for G over F.

Suppose now that $G = N \rtimes H$. Assume that as an H module, N has the form $N_1 \oplus \cdots \oplus N_r$. Let T'/R' be a generic Galois extension for H over F, and suppose S_i/R_i are generic ring extensions of T'/R' for the groups $N_i \rtimes H$. The next result shows how to construct a generic extension of T'/R' for $N \rtimes H$.

THEOREM 3.9. *Let N, N_i, T'/R', and S_i/R_i all be as above. Then there is a generic ring extension of T'/R' for $N \rtimes H$.*

Proof. Set $R = R_1 \otimes_{R'} \cdots \otimes_{R'} R_r$ and $T = T' \otimes_{R'} T$. Consider $T_i = T' \otimes_{R'} R_i$ as a subring of T and set $S_i' = S_i \otimes_{T_i} T$. Finally, set $S = S_1' \otimes_T \cdots \otimes_T S_r'$. As $S_i' = S_i \otimes_{R_i} R$, S_i'/R is Galois with group $N_i \rtimes H$. Also, S/T is Galois with group N. It now follows, using 0.1, that S/R is Galois with group $N \rtimes H$.

That R satisfies 3.8(a) is clear. It remains to show that S/R satisfies 3.8(c). But suppose K is a feld and L/K a Galois extension with group $N \rtimes H$. $L = L_1 \otimes_{K'} \cdots \otimes_{K'} L_r$, where K' is the fixed ring of N and L_i/K has group $N_i \rtimes H$. Choose $\varphi': R' \to K$ realizing K'/K, and extend it to $\varphi_i: R_i \to K$ realizing L_i/K. By tensoring the φ_i's we have $\varphi: R \to K$ realizing L/K.

Q.E.D.

The next theorem, a generalization of 3.7, is just an application of 3.9.

THEOREM 3.10. *Let N be a finite abelian group of exponent n and H a finite group acting on N. Assume N is a direct sum, as H modules, of cyclic groups N_i of prime power order, such that the image of H in* $\mathrm{Aut}(N_i)$ *is cyclic. Finally assume n is prime to the characteristic of F and $\rho(n) \in F$. Then there is a generic Galois extension for $N \rtimes H$ over F.*

COROLLARY 3.11. *Suppose N is a cyclic group of order n and H a finite group acting on N. Assume n is prime to the characteristic of F and $\rho(n) \in F$. Then if n is odd, or if n is even but H has cyclic image in* $\mathrm{Aut}(N)$, *$N \rtimes H$ has a generic Galois extension over F.*

This section will end with some observations about constructing Galois extensions which are not generic. The idea is that the constructions of this section apply in the case when no generic property is being assumed. Instead we consider regular extensions, which we are about to define. Let F be an infinite field. An F regular Galois extension with group G is an extension of fields L/K such that K is a purely transcendental extension of F, L/K is Galois with group G, and if $\bar{F}$ is the algebraic closure of F, then L and $\bar{F}$ are linearly disjoint. If F is a Hilbertian field (e.g. [7, p. 141]), then the existence of an F regular G extension implies that G appears as the Galois group of F'/F for some field F'. Many of the constructions of this and previous sections can be easily translated to be about regular extensions. The following theorem is the result.

THEOREM 3.12. *Let F be an infinite field, and G, N finite groups.*

(a) *If S/R is generic for G over F, and S is a domain, then there is a F regular extension with group G.*

(b) *If there are F regular extensions for G and N then there is one for $G \oplus N$.*

(c) *If N is an abelian group, then F has a regular extension with group N (use tractable extensions).*

(d) *If there are F regular extensions for G and N, there is one for $N \wr G$.*

(e) *If F has a regular extension for G and N is a normal subgroup of G, then F has regular extension for G/N.*

(f) *Suppose N is abelian, and G acts on N. Assume that there is an F regular extension for G. Then there is an F regular extension for $N \rtimes G$.*

When F is Hilbertian, 3.12 yields ways of constructing Galois groups over F. It is therefore of interest to determine exactly the class of groups constructible by iterating (b), (c), (d), (e), and (f) above. This author does not know the answer to this question.

4. A Construction on Characteristic p

In this section we turn to fields of prime characteristic p and extensions of p groups. It is in this context that we can construct generic Galois extensions for nonsplit group extensions. For example, we can construct a generic extension for any p group (this was first done in [12]). The reader should note that our argument is not far removed from the original construction by Witt of p groups over fields of characteristic p [15].

Our procedure will be an inductive one. We begin by fixing some notation. Let F be a finite field of characteristic $p \neq 0$, and G a finite group. Suppose that G has a normal subgroup $C \subset G$ with the following properties. First of all, C is an abelian group of exponent p. Secondly, if $G' = G/C$, then the induced map $\psi: G' \to \mathrm{Aut}(C)$ has abelian image H which is either trivial or of order prime to p. Let n be the exponent of H.

THEOREM 4.1. *Suppose $\rho(n) \in F$, and S/R is a generic extension for G' over F. Further assume that if H is nontrivial, the induced subextension S'/R with group H has the form $R\{y^m - a\} \otimes_R \cdots \otimes_R R\{y^r - b\}$. Then there is a generic extension for G over F.*

Before proving this theorem, we record a consequence which was announced in [11].

COROLLARY 4.2. *Let H be an abelian group of exponent n, prime to p. Let P be a p group and $\psi: H \to \mathrm{Aut}(P)$ a homomorphism. Then if $\rho(n) \in F$, there is a generic extension for $P \rtimes_\psi H$ over F.*

Proof of the corollary. Let $Z \subset P$ consist of those elements of the center of P of order dividing p. Of course, Z is mapped into itself by the action of H. Using induction, we may assume $P/Z \rtimes_\psi H$ has a generic Galois extension over F. We are done by 4.1.

The rest of this section will be taken up by the proof of 4.1. Let C, G, G', ψ, and H be as in that theorem. For any prime power q, denote by $F(q)$ the field of q elements. Since C has an action by H and has exponent p, we can consider C as a module over the group algebra $F(p)[H]$. C is a direct sum of irreducible $F(p)[H]$ modules. So suppose $C = C_1 \oplus C_2$, where $C_i \neq 0$. Set $G'' = G/C_1$, and let H' be the image of H in $\mathrm{Aut}(C_2)$. Consider C_2 as a subgroup of G''. Then C_2, G'', G', and H' satisfy the conditions of 4.1. By induction on the order of C we conclude that there is a generic extension for G''. Considering C_2, G, and G'', we repeat the above argument to yield a generic extension for G. In other words, we may assume C is an irreducible $F(p)[H]$ module. If H is trivial, this of course means that C has order p.

The irreducible modules over $F(p)[H]$ can be described as follows. Let q be the order of C. There is an isomorphism $\eta: C \cong F(q)^*$ and a

homomorphism $\delta: H \to F(q)^*$ such that the action of H on C corresponds to the multipication action of $F(q)^*$ on $F(q)^+$. Since $H \to \mathrm{Aut}(C)$ is an injection, δ is an injection and H must be cyclic. Hence n, the exponent of H, is also the order of H. Assume H is nontrivial. As C is irreducible, $\delta(H)$ generates $F(q)$. But $\delta(H)$ is generated by $\rho(n)$ and $\rho(n) \in F$. Thus $F(q) \subseteq F$. If H is trivial $q = p$ and, of course, $F(p) \subseteq F$.

The obstruction to the splitting of $G \to G/C = G'$ is a two cocycle of G' on C defined as follows. Choose, for each $\sigma \in G'$, a preimage $z_\sigma \in G$. Define $c(\sigma, \sigma') = z_\sigma z_{\sigma'} z_{\sigma\sigma'}^{-1} \in C$. $c(\sigma, \sigma')$ is the cocycle mentioned above, and $c(\sigma, \sigma')$ is a coboundary if and only if the map $G \to G'$ splits. Note that ψ, $c(\sigma, \sigma')$, C and C' completely define G.

Suppose R is an F algebra and S'/R is Galois with group G'. There is a subextension S''/R with group H. Of course, $S'' = R$ if H is trivial. Assume there is a unit $\alpha \in S''$ such that

$$\sigma(\alpha) = \alpha\delta(\sigma) \qquad \text{for all} \quad \sigma \in H. \tag{1}$$

We take $\alpha = 1$ if H is trivial. Setting $v = \alpha^n \in R$, then $S' \cong R\{y^n - v\}$. Denote by u the element $\alpha^{-(q-1)}$. Define an embedding η': $C \to (S')^+$ by $\eta'(\tau) = \eta(\tau)\,\alpha$. We note that η' preserves the action of G' on C and S'. In particular, $\eta'(c(\sigma, \sigma'))$ is a two cocycle of G' on $(S')^+$. Since $H^2(G', (S')^+) = (0)$, there is a map d: $G' \to S'$ such that

$$d(\sigma) + \sigma(d(\sigma')) - d(\sigma\sigma') = \eta'(c(\sigma, \sigma')). \tag{2}$$

Every element of $\eta'(C)$ satisfies the equation $y^q - uy = 0$. Thus setting $\mathscr{P}_u$ to be the function $\mathscr{P}_u(y) = y^q - uy$, we have that $\mathscr{P}_u(d)$ is a one cocycle. Again, $H^1(G', (S')^+) = (0)$ and so there is an $s \in S'$ such that

$$\sigma(s) - s = \mathscr{P}_u(d(\sigma)) \qquad \text{for all} \quad \sigma \in G'. \tag{3}$$

Of course, if $s' \in S'$ also satisfies (3) then $s - s' \in R$. This element s is used to construct an extension of S' with Galois group G over R.

LEMMA 4.3. *Set $S = S'\{y^q - uy - s\}$. Then S/R is Galois with group G. Conversely, suppose $R = K$ is a field, and $L \supseteq L' \supseteq K$ are such that L/K is Galois with group G and L' is the fixed ring of C. Let $\alpha \in L'$ be any unit satisfying* (1) *and d: $G' \to L'$ any map satisfying* (2). *Then $L \cong L'\{y^q - uy - s'\}$ for some $s' \in L'$ satisfying* (3).

Proof. Let $\beta \in S$ be a canonical generator for S/S'. Define an action of G on S as follows. Let G act on S' via the map $G \to G'$. Every element of G can be written uniquely as τz_σ for $\sigma \in C$, $\sigma \in G'$. Define $\tau z_\sigma(\beta) = \beta + d(\sigma) + \eta'(\tau)$. The relations (1), (2), and (3) show that this defines an action of G on S. In order to prove that S/R is Galois with group G, it

suffices to prove that S/S' is Galois with group C. Set $\beta' = \beta\alpha^{-1}$ and note that $\tau(\beta') = \beta' + \eta(\tau)$. Also, $(\beta')^q - \beta' \in S'$. S is then isomorphic to $S'\{y^q - y - a\}$ for some $a \in S'$, and an easy exercise shows that this is Galois as claimed.

As for the converse, let α and $d(\sigma)$ be as given. Set $v = \alpha^n$ and $u = \alpha^{-(q-1)}$. Use α to define η' as before. Consider the map $f(\tau z_\sigma) = d(\sigma) + \eta'(\tau)$. Calculate that $f(\tau z_\sigma) + \tau z_\sigma(f(\tau' z_{\sigma'})) - f(\tau z_\sigma \tau' z_{\sigma'}) = d(\sigma) + \eta'(\tau) + \sigma(d(\sigma')) + \delta(\sigma)\eta'(\tau) - [d(\sigma\sigma') + \eta'(\tau) + \delta(\sigma)\eta'(\tau') + \eta'(c(\sigma, \sigma'))] = 0$. Since $H^1(G, L^+) = 0$, there must be a $\beta \in L$ such that $\tau z_\sigma(\beta) = \beta + d(\sigma) + \eta'(\tau)$. Set $s' = \beta^q - u\beta$. Then s' is fixed by all τ and so $s' \in L'$. In addition, $\sigma(s') - s' = \mathscr{P}_u(d(\sigma))$. Since β generates L over L', this finishes 4.3. Q.E.D.

All is now ready for the proof of Theorem 4.1. Let S'/R' be a generic Galois extension for G' over F, as in the statement of 4.1. Because of the assumptions, there is a unit $\alpha \in S'$ such that (1) holds. Choose d: $G'(S')^+$ and $s \in S'$ such that (2) and (3) hold. Set $R = R'[x]$ and $S'' = S'[x]$, where G' acts trivially on x. Set $v = \alpha^n$, $u = \alpha^{-(q-1)}$, and $S = S'\{y^q - uy - (s + x)\}$.

THEOREM 4.4. *S/R is generic for G over F.*

Proof. Let L/K be Galois with group G and $L' \subset L$ the fixed ring of C. Let φ': $R' \to K$ be any specialization which realizes L'/K. Denote by φ'' the induced map $S' \to L'$. Now $\varphi''(\alpha)$, $\varphi''(d(\sigma))$, and $\varphi''(s)$ satisfy (1), (2), and (3). $L \cong L'\{y^q - uy - s'\}$, where s' satisfies (3). Hence $s' = \varphi''(s) + a$, $a \in K$. Extend φ' to R by setting $\varphi(x) = a$. φ now clearly realizes L/K.

Q.E.D.

5. NOETHER'S CONSTRUCTION AND THE GRUNWALD–WANG THEOREM

In this section we collect a series of results relating generic Galois extensions and some other questions and results in the theory of fields. We begin with Noether's construction and show that an affirmative answer to Noether's Problem implies the existence of a generic Galois extension. We next show that the existence of a generic Galois extension allows one to "lift" Galois extensions over a local ring. In fact, we prove that this lifting property is necessary and sufficient for the existence of a generic extension. Finally, we investigate a topic which in number theory is called the Grunwald–Wang theorem. Namely, we show that the existence of a generic Galois extension implies that one can "pullback" Galois extensions from complete fields to dense subfields. We can prove our theorems in the context of arbitrary real valued valuations and arbitrary groups with a generic extension. Included in our results is an elementary proof of a large chunk of

the classical Grunwald–Wang theorem about global fields, including enough to prove the cyclicity of division algebras over global fields. We also provide some results concerning the more general problem of defining extensions with specified decomposition groups and completions.

As promised we will start with Noether's construction. Let F be an infinite field and G a finite group. Form the rational function field $L = F(x_\sigma \mid \sigma \in G)$, where the x_σ's are indeterminants. Let G act on L by letting G act trivially on F and by setting $\sigma(x_\tau) = x_{\sigma\tau}$. If L^G is the fixed field of L under G, Noether's problem is whether L^G is rational over F. Swan [14] first showed that when $G = C_{47}$ and F is the field, Q, of rational numbers, then L^G is not rational over Q. Lenstra has classified [8] the abelian groups G such that L^G is rational over F. We will give an elementary proof that when $G = C_8$ and $F = Q$, L^G is not rational over Q. Only the method is new; this fact is in [8].

When Noether first made this construction [10], one motivation was to construct generic polynomials with given Galois group G. Kuyk in [6] has elaborated on this theme. We now present a theorem which is really a slight modification of Theorem 1 of [6]. We give a proof of this theorem for completeness sake and because our theorem is given in terms of our new terminology.

THEOREM 5.1. *Let F, L, and G be as above, with F infinite. If L^G is rational over F, then there is a generic Galois extension for G over F.*

Remark. The converse of 5.1 is false. C_{47} has a generic extension over Q, but Swan showed Noether's Problem false for this groups.

Proof. Suppose $a_1, \ldots, a_n \in L^G$ are a transcendence base for L^G over F. Set $R_1 = F[a_1, \ldots, a_n] \subset L^G$. By 0.2(c), there is an $s \in R_1$ and an algebra $S \supseteq R = R_1(1/s)$ such that S/R is Galois with group G and $S \otimes_R L^G \cong L$. We claim that S/R is a generic Galois extension for G over F.

Write $a_i = f_i/g_i$, where the f's and g's are polynomials in the x_σ's. Let $e_1, \ldots, e_m$ generate S as a module over R and write $e_j = h_j/k_j$, where the h's and k's are also polynomials in the x_σ's. Then if t is the product of all the g's and k's, we have that $S \subset F[x_\sigma \mid \sigma \in G](1/t)$.

Suppose K is a field, $K \supseteq F$, and L/K is Galois with group G. In order to realize L/K it suffices to define $\varphi: S \to L$ such that φ preserves G, and $\varphi(S)K = L$. In order to define such at φ, it suffices to define $\varphi: F[x_\sigma \mid \sigma \in G] \to L$ such that $\sigma(\varphi(x_1)) = \varphi(x_\sigma)$, the $\varphi(x_\sigma)$'s are a basis for L over K, and t evaluated at the $\varphi(x_\sigma)$'s is a unit. Thus, in fact, it suffices to prove the following lemma.

LEMMA 5.2. *Let L/K be as above, and t any polynomial in $F[x_\sigma \mid \sigma \in G]$. Then there is a $z \in L$ such that $\{\sigma(z) \mid \sigma \in G\}$ is a normal basis for L over K and such that t evaluated at the $\sigma(z)$'s is a unit.*

Proof. Letting G act on $F[x_\sigma \mid \sigma \in G]$, we form $t' = \prod_{\sigma \in G} \sigma(t)$. t' has the property that $t'(z, \sigma(z),..., \sigma'(z))$ is always in K, and if $t'(z, \sigma(z),..., \sigma'(z)) \neq 0$, then $t(z, \sigma(z),..., \sigma'(z))$ is a unit. In other words, we may assume $t(z, \sigma(z),..., \sigma'(z))$ is always in K and find a z such that $t(z, \sigma(z),..., \sigma'(z)) \neq 0$. Let $e_1,..., e_n$ be a basis for L over K, with $e_1 = 1$. Formally write $z = y_1 e_1 + \cdots + y_n e_n$, where the y's are meant to take values in K. $\sigma(z)$ is then of the form $l_1 e_1 + \cdots + l_n e_n$, where each l_i is a linear polynomial in the y's. Taking determinants, we have a polynomial $f(y_1,..., y_n)$ such that $f(b_1,..., b_n) \neq 0$ if and only if $z = b_1 e_1 + \cdots + b_n e_n$ generates a normal basis. Since some normal basis exists, f is a nonzero polynomial. Consider $t(z, \sigma(z),..., \sigma'(z))$. Substituting the above expressions for the $\sigma(z)$'s, we have that $t(z, \sigma(z),..., \sigma'(z)) = f'(y_1,..., y_n)$ for some polynomial f' with coefficients in K. By the algebraic independence of the σ's, f' is nonzero. Thus it suffices to find $a_1,..., a_n \in K$ such that $f'(a_1,..., a_n) f(a_1,..., a_n) \neq 0$. This is always possible because K is infinite. Lemma 5.2 and Theorem 5.1 have been proven.

Q.E.D.

Theorem 5.1 supplies another tool for the construction of generic Galois extensions, as there are some positive results that show that for certain G, L^G is rational. For example, one can show that the symmetric group S_n has a generic Galois extension over any infinite field F.

The existence of a generic Galois extension for G is now going to be shown to be equivalent to a lifting property for G Galois extensions over local rings. This lifting problem was the beginning of this author's interest in this material.

Let G be a finite group and F a field. We say that (G, F) has the lifting property if for all semi-local F algebras T with Jacobson radical M, and all G Galois extensions L/K with $K = T/M$, there is a Galois extension T'/T such that $T' \otimes_T T/M \cong L$. The connection between the lifting property and generic extensions is as follows.

THEOREM 5.3. *The pair (G, F) has a generic Galois extension if and only if the same pair has the lifting property.*

Proof. First assume (G, F) has a generic Galois extension S/R. Write $R = F[x_1,..., x_n](1/s)$. Suppose T is a semi-local F algebra with Jacobson radical M. Then $K = T/M$ is a finite direct sum of fields. Assume L/K is Galois with group G. By Proposition 1.4, there is an F algebra map $\varphi: R \to K$ such that $S \otimes_\varphi K \cong L$. Write $a_i = \varphi(x_i)$ and choose a preimage $b_i \in T$ for each a_i. Define $\varphi': R \to T$ by setting $\varphi'(x_i) = b_i$. φ' is well defined because $\varphi'(s) \in T$ is a preimage for $\varphi(s)$ and so is invertible. Setting $T' = S \otimes_{\varphi'} T$, we have that $T' \otimes_T K \cong L$. Thus (G, F) has the lifting property.

Conversely, assume (G, F) has the lifting property. Define $L = F(x_\sigma \mid \sigma \in G)$ as in 5.1, and consider the field L^G. By 0.2(c), there is a domain

$R' \subset L^G$ and a G Galois extension S'/R' such that R' is finitely generated over F as an algebra, and such that $L \cong S' \otimes_{R'} L^G$. Since S' is finitely generated over R, there is a $0 \neq t \in F[x_\sigma \mid \sigma \in G]$ such that $S' \subset F[x_\sigma \mid \sigma \in G](1/t)$. Hence, arguing exactly as in 5.1, if K is a field and L/K is G Galois, then there is a φ': $R' \to K$ such that $L \cong S' \otimes_{\varphi'} K$. Furthermore, if $0 \neq s \in R'$, we can choose φ' such that $\varphi'(x) \neq 0$.

The difficulty is, of course, that R' is not necessarily of the form $F[x_1,..., x_n](1/s)$. Suppose $a_1,..., a_n$ generate R' over F. Let $F[z_1,..., z_n]$ be the polynomial ring and define: $F[z_1,..., z_n] \to R'$ by $\varphi(z_i) = a_i$. Localize $F[z_1,..., z_n]$ at the kernel of φ to get a local ring T and a homomorphism φ: $T \to K'$, where K' is the field of quotients of R'. Using the lifting property, these is a G Galois extension T'/T such that $T' \otimes_\varphi K' \cong S' \otimes_{R'} K'$. By 0.2(c) again, there is an $s \in F[z_1,..., z_n]$ with $1/s \in T$ and a ring S'' such that if $R'' = F[z_1,..., z_n](1/s)$, then S''/R'' is G Galois and $T' \cong S'' \otimes_{R''} T$. Set $s' = \varphi(s)$. φ induces a surjection φ: $R'' \to R'(1/s')$. It may not be true that $S'' \otimes R'(1/s') \cong S' \otimes_{R'} R'(1/s')$, but these Galois extensions are isomorphic after tensoring with K'. This isomorphism can be defined using finitely many elements of K. It follows that there is a $t' \in R'$ such that $S'' \otimes_\varphi R'(1/s't') \cong S' \otimes_{R'} R'(1/s't')$. Let $t \in F[z_1,..., z_n]$ be a preimage of t' and set $R = R''(1/t)$, $S = S'' \otimes_{R''} R$. Then we have φ: $R \to R'(1/s't')$ and $S \otimes_R R'(1/s't') \cong S' \otimes_{R'} R'(1/s't')$.

We claim that S/R is a generic Galois extension for G over F. 1.1(a) and 1.1(b) are clear. As for 1.1(c), suppose L/K is G Galois. There is a φ': $R' \to K$ such that $\varphi'(s't') \neq 0$ and φ' realizes L/K. φ' extends to $R'(1/s't')$ and $\varphi' \circ \varphi$ realizes L/K. Q.E.D.

We have proved 5.3. But before leaving the above construction, note the following fact. Suppose S/R to be the generic Galois extension constructed above, and φ: $R \to R'(1/s't')$ is the map defined in that construction. If $u \in R$ and $\varphi(u) \neq 0$, then our argument above also shows that $S(1/u)/R(1/u)$ is a generic Galois extension for G over F. In fact, 1.1(a) and (b) are clear. If L/K is G Galois then choose φ': $R' \to K$ such that φ' realizes L/K and $\varphi'(\varphi(u)\, s't') \neq 0$. Then $\varphi' \circ \varphi$ extends to $R(1/u)$ and realizes L/K. This verifies 1.1(c).

The above observation can be used to fill in a gap left in the arguments of Section 3. Namely, we observe that if S/R is a generic Galois extension for G over F, then one may assume S is free as an R module.

COROLLARY 5.4. *Let G be a group and F a field such that there is a generic Galois extension for G over F. Then some such extension S/R has the additional property that S is free as an R module.*

Proof. If there is a generic extension for G over F, then the pair (G, F) has the lifting property. Using the lifting property, it was shown in 5.3 how

to construct a generic extension for G over F. Let S/R, S'/R', and $\varphi: R \to R'(1/s't')$ be as in that construction. Recall that $S \otimes_\varphi R'(1/s't') \cong S'(1/s't')$. Since $S'(1/s't')$ is projective as a module over $R'(1/s't')$, there is a $u \in R$ such that $S(1/u)$ is free as a module over $R(1/u)$ and $\varphi(u) \neq 0$. We observed just after 5.3 that $S(1/u)\, R(1/u)$ is a generic Galois extension for G over F. Q.E.D.

Let us now turn to developing a generalized Grunwald–Wang theorem. We begin with some facts about valued fields. Let K be a field complete under a nonarchimedean real valued valuation v. Extend v to the polynomial ring $K[y]$ by defining $v(f(y))$ to be the maximum among $v(c_i)$, where the c_i are the coefficients of $f(y)$. Also note that v extends to the algebraic closure, $\bar{K}$, of K. Krasner's lemma (e.g. [1, pp. 44–45]) implies that "close" polynomials in $K[y]$ define isomorphic extensions of K. But because the common references only handle irreducible polynomials, we state and prove the following variant of this well-known result.

LEMMA 5.5. *Let $f(y) \in K[y]$ be a separable polynomial of degree r. There exists a real number $\varepsilon > 0$ such that for any $g(y) \in K[y]$ of degree r, if $v(f(y) - g(y)) < \varepsilon$ then $K\{f(y))\} \cong K\{g(y)\}$.*

Proof. Let $\alpha_1, \ldots, \alpha_r$ be the roots of $f(y)$ in $\bar{K}$. In the usual proof of Krasner's lemma it is shown that for any $\delta > 0$, there is an $\varepsilon > 0$ such that for any $g(y) \in K[y]$ satisfying $v(f(y) - g(y)) < \varepsilon$, and any root α_i of $f(y)$, there is a root β of $g(y)$ such that $v(\alpha_i - \beta) < \delta$. Choose such an ε for $\delta = \min v(\alpha_i - \alpha_j)$. For a $g(y)$ satisfying $v(f(y) - g(y)) < \varepsilon$, denote by β_i a root of $g(y)$ such that $v(\alpha_i - \beta_i) < \delta$. Krasner's lemma proper shows that the field $K(\beta_i)$ contains $K(\alpha_i)$. But writing $\beta_i - \beta_j = (\beta_i - \alpha_i) + (\alpha_i - \alpha_j) + (\alpha_j - \beta_j)$, and noting that $v(\beta_i - \alpha_i)$ and $v(\alpha_j - \beta_j)$ are less than $v(\alpha_i - \alpha_j)$, we have that $v(\beta_i - \beta_j) = v(\alpha_i - \alpha_j)$. Hence all the β_i's are distinct and are all the roots of g. Krasner's lemma applied again show that $K(\beta_i) = K(\alpha_i)$. Since $K\{f(y)\}$ and $K\{g(y)\}$ are direct sums of fields defined by the irreducible factors of $f(y)$ and $g(y)$, respectively, the lemma is clear. Q.E.D.

One can now take Lemma 5.5 and generalize it from polynomials to specializations in the sense of this paper. More precisely, let G be a finite group and S/R a G Galois extension such that $R = F[x_1, \ldots, x_n](1/t)$. Let $\varphi: R \to K$ be an F algebra map and set $L = S \otimes_\varphi K$. We next show that specializations of R "close" to φ yield isomorphic extensions. Set $a_i = \varphi(x_i)$.

LEMMA 5.6. *There is an $\delta > 0$ such that if $b_i \in K$ satisfy $v(b_i - a_i) < \delta$ for all i, then setting $\varphi'(x_i) = b_i$ defines a specialization $\varphi': R \to K$ such that $L \cong S \otimes_{\varphi'} K$ as Galois extensions.*

Proof. By considering $S \otimes_F K/R \otimes_F K$, we may assume $F = K$. Let r be

the order of G. As K is finite and L a direct sum of fields separable over K, one can write $L \cong K\{f(y)\}$, where f is a separable monic polynomial. Set $\alpha \in L$ to be a canonical generator. Letting φ also denote the induced map $\varphi: S \to L$, we have that since $F = K$, φ is surjective. Choose $\beta \in S$ such that $\varphi(\beta) = \alpha$. Set $p(y) = y^r - \mathrm{tr}(\beta)\, y^{r-1} + \cdots + n(\beta) \in R[y]$ to be the characteristic polynomial of β. As $p(\beta) = 0$, the image of $p(y)$ under φ is exactly $f(y)$. Now $1, \beta, \ldots, \beta^{r-1}$ may not be a basis of S over R, but there is an $s \in R$ such that $\varphi(s) \neq 0$ and $1, \beta, \ldots, \beta^{r-1}$ is a basis for $S(1/s)$ over $R(1/s)$.

Thus to find δ, we first choose δ so small that if $v(b_i - a_i) < \delta$ for all i, then $t(b_1, \ldots, b_n) \neq 0$ and $s(b_1, \ldots, b_n) \neq 0$. Setting $\varphi'(x_i) = b_i$, we have defined a map $\varphi': R(1/s) \to K$. φ' induces a map $\varphi': S(1/s) \to L$. Write $p(y) = y^r + (g_1/t^m)\, y^{r-1} + \cdots + g_r/t^m$, where g_i, $t \in F[x_1, \ldots, x_n]$. Then $g_i(a_1, \ldots, a_n)/t(a_1, \ldots, a_n)^m$ is the ith coefficient, c_i, of $f(y)$. hoose $\varepsilon > 0$ as in Lemma 5.5. We can make δ so small that if $v(b_i - a_i) < \delta$, then $v((g_i(b_i, \ldots, b_n)/t(b_1, \ldots, b_n)) - c_i) < \varepsilon$. We have, therefore, that $S \otimes_{\varphi'} K \cong L$ as K algebras. To prove the lemma, we only need add that $S \otimes_{\varphi'} K$ and L have the same decomposition groups (see 0.4). So let $H \subseteq G$ be a decomposition group of G and L. Set $L' = L^H$ and note that L' has K as a direct summand. Denote by T the fixed ring S^H. Arguing as above, we can choose δ so small that $T \otimes_{\varphi'} K \cong L'$. Hence $T \otimes_{\varphi'} K$ has K as a direct summand. One condludes that using this δ, H is a decomposition group for $S \otimes_{\varphi'} K$. Q.E.D.

For completeness sake, we derive the result corresponding to 5.6 for complete archimedean real valued valuations. Such fields are the real or complex field, denoted $\mathbf{R}$ or $\mathbf{C}$, respectively (e.g. [1, p. 24]). So let K be R or C, and v the valuation. Extend v to $K[y]$ by setting $v(c_0 + c_1 y + \cdots + c_r y^r) = v(c_0) + \cdots + v(c_r)$.

LEMMA 5.7. *Suppose K is $\mathbf{R}$ or $\mathbf{C}$, and v is the corresponding valuation.*

(a) *Let $f(y) \in K[y]$ by a separable polynomial. There is an $\varepsilon > 0$, such that if $v(f(y) - g(y)) < \varepsilon$ and $g(y) \in K[y]$ has degree equal to f's then $K\{f(y)\} \cong K\{g(y)\}$.*

(b) *Let S/R be Galois with group G and $R = F[x_1, \ldots, x_n](1/t)$. Suppose $\varphi: R \to K$ is an F algebra map and $L = S \otimes_\varphi K$. Then there is an $\varepsilon > 0$ such that if $v(b_i - \varphi(x_i) < \varepsilon$ then setting $\varphi'(x_i) = b_i$ defines a specialization $\varphi': R \to K$ and $L \cong S \otimes_{\varphi'} K$.*

Proof. Of course, both (a) and (b) are trivial if $K = \mathbf{C}$. Also, that (a) implies (b) is proved exactly as in 5.6. Thus we may assume $K = \mathbf{R}$ and prove (a). But $K\{f(y)\} \cong K\{g(y)\}$ if and only if f and g have equal numbers of real roots. The lemma is now an easy exercise. Q.E.D.

We have prepared the ground for the next series of theorems which should be thought of as generalizations of the Grunwald–Wang theorem. Let K be a

field with inequivalent real valued valuations $v_1,\ldots,v_m$. Denote by K_i the completion of K with respect to v_i. The idea of the next theorem is that using a generic Galois extension, one can simultaneously approximate extensions of the K_i by a single extension of K. In part (b) below, we observe that the exact same argument applies to approximating tractable extensions.

THEOREM 5.8. (a) *Let S/R be a generic Galois extension for G over $F \subseteq K$. Assume for each i, there is a G Galois extension L_i/K_i. Then there is a G Galois extension L/K such that $L \otimes_K K_i \cong L_i$.*

(b) *Suppose G is cyclic of 2 power order and there are tractable G Galois extensions L_i/K_i. Then there is a tractable G Galois extension L/K such that $L \otimes_K K_i \cong L_i$.*

Proof. We will prove part (a), as (b) is exactly the same. So assume the situation is as in (a). Take $\varphi_i\colon R \to K_i$ which realize L_i/K_i. Set $a_{ij} = \varphi_i(x_j)$. Choose an ε_i for each φ_i and K_i in Lemmas 5.6 and 5.7. By taking the minimum, we can assume all the ε_i's are equal to a single ε. By the weak approximation theorem (e.g. [1, p. 10]), there are $b_j \in K$ such that $v_i(b_j - a_{ij}) < \varepsilon$ for all i, j. The map $\varphi\colon R \to K$ defined by $\varphi(x_i) = b_i$ is well defined and if $L = S \otimes_\varphi K$, then $L \otimes_K K_i \cong L_i$. Q.E.D.

In our context the L_i above need not be fields. Of course, if even one L_i is a field, then L must be a field. Allowing the L_i not to be fields means that 5.8 contains within itself the seemingly more general embedding problem, as we now show. Suppose L/K is a G Galois extension of fields. Set $L_i = L \otimes_K K_i$. If e is a primitive idempotent of L_i, then $H = \{\sigma \in G \mid \sigma(e) = e\}$ has been defined as a decomposition group of G and L_i. In this context, it is called a decomposition group of G at v_i. Recall that any conjugate of H is a decomposition group, and that all decomposition groups are conjugate. Finally, eL_i is the completion of L at the extension of v_i (e.g. [1, p. 223]).

The next result shows that extensions L/K can be chosen with prescribed decomposition groups and completions. The theorem is really a corollary of 5.8, but here we keep straight what is and is not a field.

THEOREM 5.9. *Suppose S/R is a generic Galois extension for G over $F \subseteq K$. Let $H_1,\ldots,H_m$ be subgroups of G such that*

(1) *Every subgroup of G containing one conjugate of each of the H_i must be all of G.*

Assume L_i/K_i are Galois extensions of fields with group H_i. Then there is a Galois extension of fields L/K with group G such that L_i is the completion of L at v_i and H_i is a corresponding decomposition group.

Proof. Let $L_i' = \mathrm{Ind}_{H_i}^G(L_i)$. Of course, H_i is a decomposition group for L_i'. Apply 5.8 to yield a G Galois extension L/K such that $L \otimes_K K_i \cong L_i'$. It remains only to show that L is a field. So suppose $0 \neq e \in L$ is a primitive idempotent. In each L_i', $e = e_1 + \cdots + e_m$, where the e_j's are primitive. Set $H = \{g \in G \mid g(e) = e\}$ and $H_i' = \{g \in G \mid g(e_1) = e_1\}$. H_i' is a conjugate of H_i. Suppose $g \in H_i'$ and $g \notin H$. Then $g(e) \neq e$. As e is primitive in L, $g(e)\, e = 0$. But since $g(e_1) = e_1$, $g(e)\, e$ has the form $e_1 + f$ for an idempotent f. This contradiction implies $H_i' \subseteq H$. Since this is true for all i, $H = G$, and $e = 1$. Q.E.D.

We remark that (1) above may be less of an obstacle to the application of 5.9 than it may at first appear. If K has more valuations than the v_i's, one is free to extend the set of H_i's in order to satisfy (1), as long as the corresponding extensions L_i/K_i can be constructed. For example, in the case G is abelian and K is a global field, one can in this way drop condition (1), as long as G is generic over K.

The above two theorems constitute the bulk of what we have been calling a generalized Grunwald–Wang theorem. Note that these results apply to all real valued valuations and all groups with a generic Galois extension. Thus the constructions of Sections 2, 3, and 4 each imply a theorem like 5.9 for the respective classes of Galois extensions. In the same way, by using 5.1 and known positive cases of Noether's problem, one can again derive Grunwald–Wang type theorems.

In the next theorem we collect the results that apply to abelian extensions.

THEOREM 5.10. *Let A be an abelian group, and K a field with real valued valuations $v_1, \ldots, v_m$ as above. Let r be the highest power of 2 dividing the exponent of A.*

(a) *Suppose K has nonzero characteristic or $K_i(\rho(r))/K_i$ is cyclic for all i. Then if L_i/K_i are A Galois extensions, there is an A Galois extension L/K such that $L \otimes_K K_i \cong L_i$.*

(b) *With the same conditions as in* (a), *let $H_1, \ldots, H_m \subset A$ be subgroups which generate A. Suppose that L_i/K_i are Galois extensions with group H_i. Then there is an a Galois extension L/K such that L_i is the completion of K at v_i and H_i is the corresponding decomposition group.*

(c) *Let A be a cyclic group, and let $n_1, \ldots, n_m$ be positive integer with greatest common divisor the order of A. Suppose each K_i has a cyclic greatest common divisor the order of A. Suppose each K_i has a cyclic extension of degree n_i. Then there is an A Galois extension of fields, L/K, such that the completion, L_i, of L at v_i has degree n_i over K_i.*

Remark. Results like Theorem 5.10(a) and (b), but a bit weaker, have previously appeared in [9] and [18], and were announced in [11]. In all

three papers, the results are limited to discrete valued fields, except for a remark in [9] about archimedean valuations on global fields.

Proof. Parts (a) and (b) follow because under these assumptions, either A is generic over K or all C_q extensions of all the K_i are tractable, where q is any power of 2 less than or equal to r. Part (c) follows using tractibility, 2.10 and 2.11. Q.E.D.

One of the most famous applications of the Grunwald–Wang theorem is to the proof that over a global field all finite dimensional division algebras are cyclic. One should note that 5.10(c) affords a proof of this part of the Grunwald–Wang theorem, and the stripped to its essentials, this proof is an elementary one.

The last theorem of this paper is the counterexample to Noether's Problem.

THEOREM 5.11. *Let A be an abelian group with an element 2 power order greater than or equal to 8. Then there cannot be a generic extension for A over the rational field Q. Also, Noether's Problem is false for A and Q.*

Proof. Let $C \subseteq A$ be a cyclic direct summand of 2 power order greater than or equal to 8. By 3.1, if there is a generic extension for A over Q, then there is one for C over Q. Let L_2/Q_2 be the unique unramified extension of the 2-adic field Q_2 with group C. In his famous counterexample to Grunwald's original theorem, Wang showed [15] that there is no C Galois extension L/Q such that $L_2 \cong L \otimes_Q Q_2$. Thus there cannot be a generic extension for C or A over Q. By 5.1, this implies that Noether's Problem is false in this case. Q.E.D.

Note added in proof. Lemma 0.4 is false as stated. The correct statement should be: Let L_1/K_2, L_2/K be G Galois extensions of the field K, with a common decomposition group $H \subseteq G$. Assume that $e_i \in L_i$ are minimal idempotents fixed by H such that $e_1 L_1 \cong e_2 L_2$ as G Galois extensions. Then $L_1 \cong L_2$ as G Galois extensions. The problem with the old version centers around the fact that if L/K is a Galois extension of fields, then the Galois group G is uniquely determined but not the action of G on L.

This change in 0.4 necessitates a change in the proof of Lemma 5.6. The second paragraph of that proof should be:

Thus to find δ, we first choose δ so small that if $v(b_i - a_i) < \delta$ for all i, then $t(b_1,..., b_n) \neq 0$ and $s(b_1,..., b_n) \neq 0$. Setting $\varphi'(x_i) = b_i$, we have defined a map $\varphi': R(1/s) \to K$. φ' induces a map $\varphi': S(1/s) \to S \otimes_{\varphi'} K$. Write $p(y) = y^r + (g_1/t^m) y^{r-1} + \cdots + g_r/t^m$, where g_i, $t \in K[x_1,..., x_n]$. Then $g_i(a_1,..., a_n)/t(a_1,..., a_n)^m$ is the ith coefficient, c_i, of $f(y)$. Choose $\varepsilon > 0$ as in Lemma 5.5. We can make δ so small that if $v(b_i - a_i) < \delta$, then $v((g_i(b_1,..., b_n)/t(b_1,..., b_n)^m) - c_i) < \varepsilon$. We have, therefore, that $S \otimes_{\varphi'} K \cong L$ as K algebras.

To finish the lemma, we must show that $S \otimes_{\varphi'} K \cong L$ as G Galois extensions. Write $L \cong L' \oplus \cdots \oplus L'$, where L' is a field. Set $e = (1, 0,..., 0) \in L$ and set $H = \{\sigma \in G | \sigma(e) = e\}$. Of course, L'/K is Galois with group H. Set $k \in K$ to be a nonzero element with $v(k) < 1$. We go back and choose $\alpha = (\alpha_1,..., \alpha_s) \in L$ with the following additional properties. Set A to be

the set of all $\gamma \in L'$ which are conjugates of some α_i and set A' to be the set of all elements of the form $\gamma + k\gamma'$, where γ, $\gamma' \in A$. If A'' is the union of A and A', we assume A'' consists of $r + r^2$ distinct elements of L' all of which generate L'. Such a choice of α is possible since K is infinite. Finally, let μ be the minimum of all $v(\gamma - \gamma')$, where γ, γ' are distinct elements of A''.

Our argument will use the following version of Krasner's lemma. Let $f(y)$ again be the characteristic polynomial of α. There is an $\eta > 0$ such that if $\alpha = (\alpha_1', ..., \alpha_s') \in L$ has characteristic polynomial $g(y)$ and $v(f(y) - g(y)) < \eta$, then for each α_1', there is a unique α_j^*, a conjugate α_j, such that $v(\alpha_i' - \alpha_j^*) < \mu$. For the purposes of this proof, we will say α_i' and α_j^* are *close* if the above condition is satisfied.

As before, let $\beta \in S$ be such that $\varphi(\beta) = \alpha$. We have already seen that for δ small enough, there is an isomorphism ψ: $S \otimes_{\varphi'} K \cong L$. By choosing δ smaller if necessary, we may assume $v(f(y) - g(y)) < \eta$ for f, g the characteristic polynomials of the following pairs of elements $\varphi'(\beta)$ and α; and $\varphi'(\beta + k\sigma(\beta))$, and $\alpha + k\sigma(\alpha)$ for all $\sigma \in H$. If $\psi(\varphi'(\beta)) = (\beta_1, ..., \beta_s)$, then we may assume ψ has been chosen such that β_i is close to α_i. Now consider, for any $\sigma \in H$, $\psi(\varphi'(\beta + k\sigma(\beta))) = (\beta_1, ..., \beta_s) + k(\alpha_1, ..., \alpha_s)$. Write $\sigma(\alpha) = (\alpha_1^*, ..., \alpha_s^*)$. Since σ fixes e, $\alpha_1^* = \sigma(\alpha_1)$. We have that $\beta_1 + k\gamma_1$ is close to some conjugate of some $\alpha_i + k\alpha_i^*$, say, $\sigma'(\alpha_i + k\alpha_i^*)$. But β_1 is close to α_1 and $k\gamma_1$ is close to some conjugate of some $k\alpha_j$, say, $k\alpha_j''$. Thus $\sigma'(\alpha_i + k\alpha_i^*)$ is close to $\alpha_1 + k\alpha_j''$. By our choice of α, this implies that $\sigma' = 1$ and $i = 1$. In other words, γ_1 is close to $\sigma(\alpha_1)$. As $\sigma(\beta_1)$ is also close to $\sigma(\alpha_1)$, we have that $\gamma_1 = \sigma(\beta_1)$. But this is true for all $\sigma \in H$, and so we conclude that $e' = \gamma^{-1}(e)$ is fixed by H and $(S \otimes_{\varphi'} K)e' \cong Le$ as H Galois extensions. Applying Lemma 0.4, this lemma is proved.

Q.E.D.

Finally, the results of Theorem 5.3 and Corollary 5.4 should be limited to *infinite* fields F. This implies that F should be assumed infinite in Section 3.

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